

INDUSTRIAL PROCESS MODELLING & SIMULATION TRAINING

2 – 4 August 2022

Part 1 : Introduction & Thermodynamic

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Topics

1.1 Introduction to Process Simulation

1.2 Getting Started with Aspen Plus

- Layouts
- User Interfaces

1.3 Thermodynamic in Aspen Plus

- Environments in Aspen Plus
- Component database
- Physical property method and selection guideline
- Property analysis

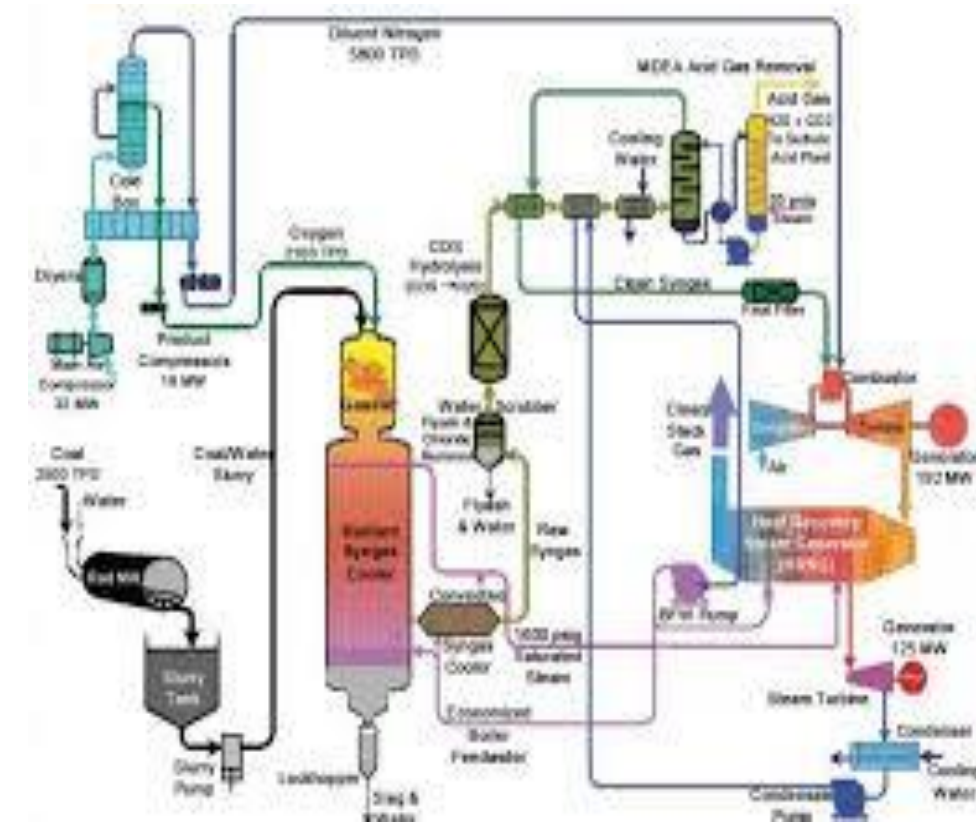
Outcome

- Differentiate between process synthesis and process analysis, and functions and applications of a process model.
- Become comfortable and familiar with the Aspen Plus Graphical User Interface (GUI)
- Develop a working knowledge of the Aspen Plus Properties Environment
- Develop a working knowledge of the Aspen Plus Simulation Environment

1.1 Introduction to Process Simulation

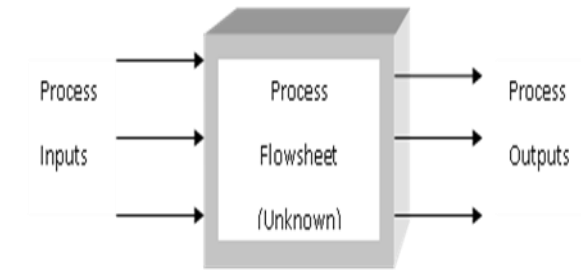
Chemical Process

- Nowadays, chemical process and processing industries span a wide range of industries including **chemical, petroleum, gas, petrochemical, pharmaceutical, food, microelectronics, metal, textile and forestry products.**
- The performance of these industries is strongly depending on their engineering (design) and engineers (operation).
- In a chemical process, the transformation of raw materials into desired products usually require more than one processing step.
- These are carried out through **reaction, separation, mixing, heating, cooling, pressure change, particle size reduction or enlargement.**
- Once individual steps have been selected, they must be interconnected to carry out the overall transformation.
- A flowsheet is then a diagrammatic representation of the processing steps with their interconnection.
- This representation can be viewed through a document called as process flow diagram (PFD).

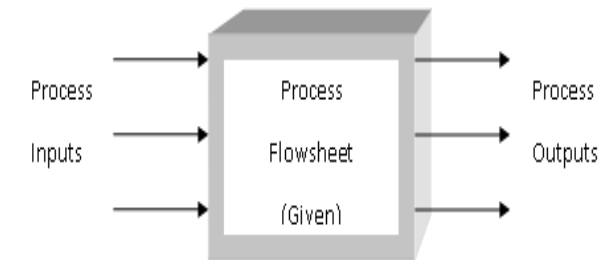


Process Synthesis vs Process Analysis

- Process synthesis: the discrete decision-making activities of conjecturing, i) which of the many available component parts one should use, and ii) how they should be interconnected to structure the optimal solution for a given design objective and problem.
- We know process inputs and outputs, to revise the structure and parameters of the flowsheet (for retrofitting design of an existing plant) or create a new flowsheet (for grass root design of a new plant). The result of process synthesis is a **flowsheet** which represents the configuration of the various pieces of equipment and their interconnection.
- Process Analysis: Involves the decomposition of the whole into its constituent elements for individual study of performance. Can be contrasted (and complemented) with process synthesis.



Process synthesis problem



Process analysis problem

Cont..

- Once an alternative is generated or a process is synthesized, its detailed characteristics (e.g flowrates, compositions, temperature and pressure) **are predicted using analysis techniques.**
- These techniques include mathematical models, empirical correlations, and **computer-aided process simulation tools.** It may involve predicting and validating performance using experiments at the lab and pilot-plant scales, and even actual runs of existing facilities.
- Hence in process analysis problems, we know the process inputs along with the structure and parameters, while we seek to determine the process outputs.

Purposes of Process Simulation

- To improve understanding of the process
- To optimize process design/operating conditions
- To perform dynamic simulation
- To design a control strategy for the process
- To train operating personnel
- To plan production or product specification

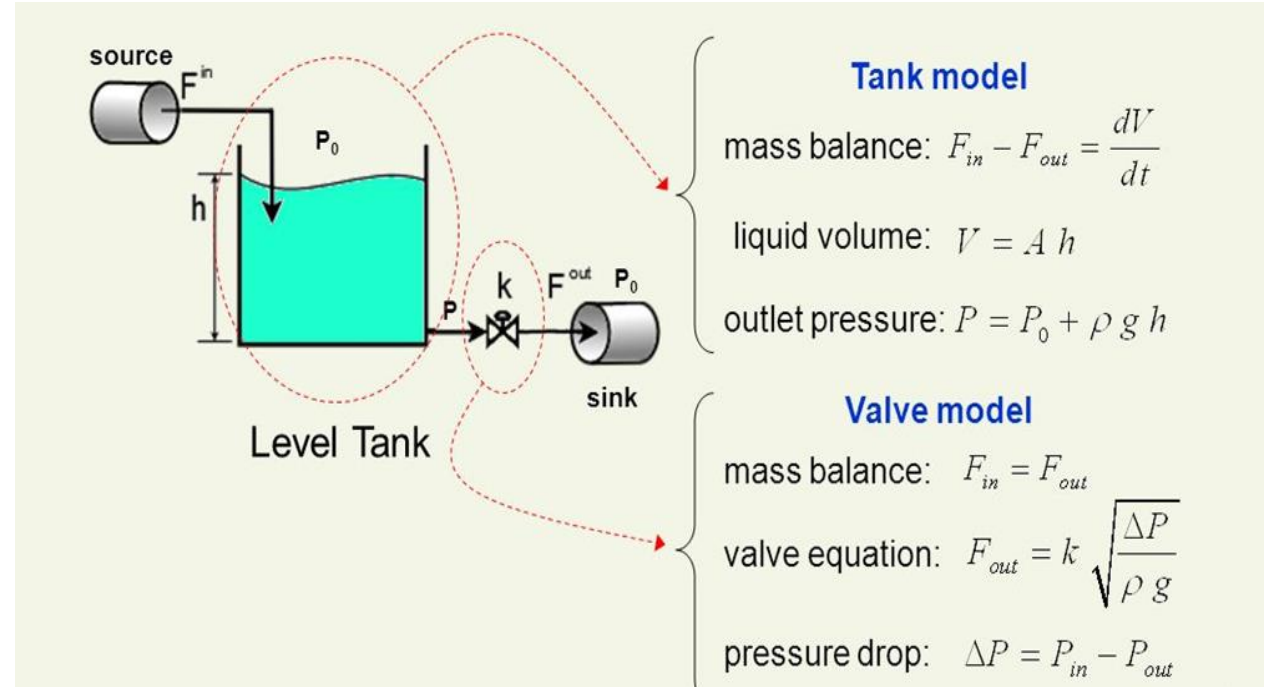


Commercial Process Simulation Software

Corporations	Software	Basic Algorithm Used
AspenTech	AspenONE Engineering (consists of Aspen Plus, Aspen HYSYS, Aspen Economic Evaluation, Aspen Exchanger Design & Rating, Aspen Energy Analyzer, Aspen Utilities Planner)	Sequential Modular (SM)
Honeywell	UniSim Design	SM
Schneider Electric	SimSci Pro/II	SM
Chemstations	ChemCAD	SM
WinSim	DESIGN II for Windows	SM
Intelligen	SuperPro Designer, SchedulePro	SM
Process Systems Enterprise	gPROMS	Equation-oriented (EO)

Equation Oriented (EO)

- Process flowsheet is treated as a set of equations.
- All model equations are gathered and solves at the same time.
- EO need good initial values.
- Able to solve much larger problems using the same computational effort.



Equation Oriented (EO)

- Very effective way of solving certain kinds of problems, such as:
 - Highly heat-integrated or recycled processes
 - Optimization
 - Model tuning via parameter estimation
 - Data reconciliation

Equation Oriented (EO)

■ Advantages

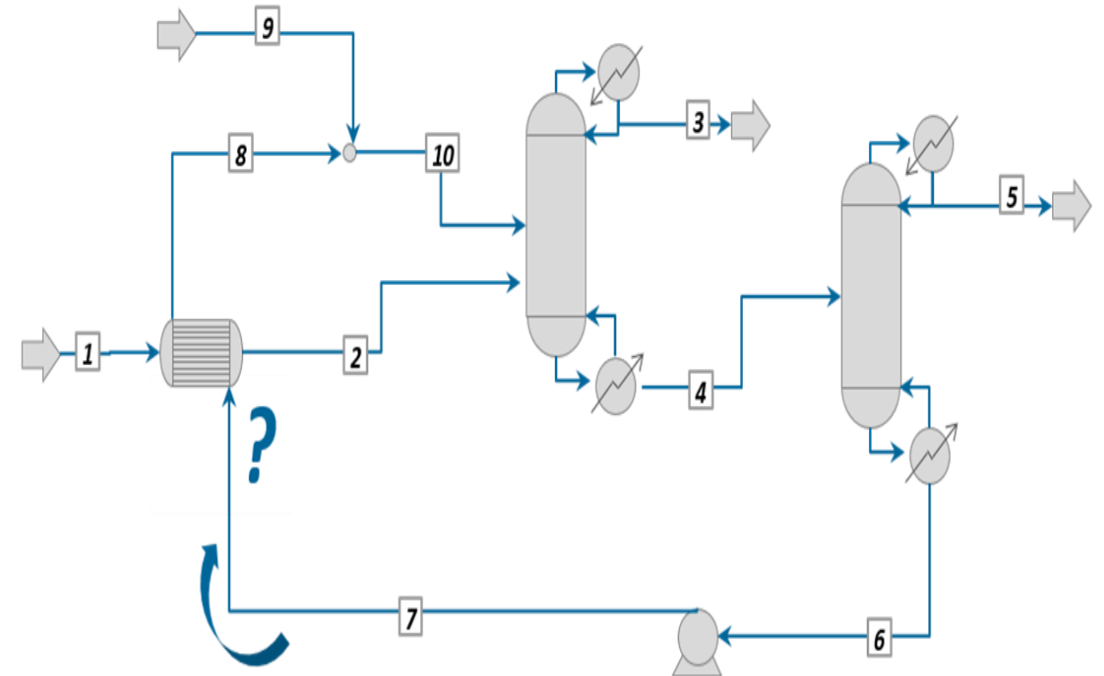
- No inherent directionality of computation – can be solved with any valid degree-of-freedom specification
- Multiple recycles do not slow down convergence – they are simply treated as any other equation
- Powerful optimization, including integer decisions
- Powerful custom modelling, with no need to program the mathematical solution
- Repeat solution is much faster, making it possible to deploy large, complex models in demanding situations such as online real-time optimization.

■ Disadvantages

- Numerical solvers may fail to find an initial solution unless good initial guesses are provided for all key variables.
- It is often difficult to provide meaningful diagnostics on failure, making debugging difficult.
- Example of EO software
 - MATLAB
 - gPROMS
 - Mathematica
 - Aspen Custom Modeler
 - Mathcad
 - Scilab (Freeware)
 - [MOSAIC](#) (Freeware)

Sequential Modular (SM)

- Process units are solved in sequence.
- A subroutine is developed for each type of equipment.
- Subroutine calculates values for the outlet streams, given the values of the inlet streams and the operating parameters for the unit
- A module library or block model is created based on the subroutine
- Modules/blocks are called from a main program in a certain sequence in order to solve all the equipment which make up the plant.





Advantages & Disadvantages of SM

■ Advantages

- Easy to use, and quick for simple calculations
- Failure is rare, and clear diagnostics are issued
- User interfaces and special solution methods can be hand-coded for each module.

■ Disadvantages

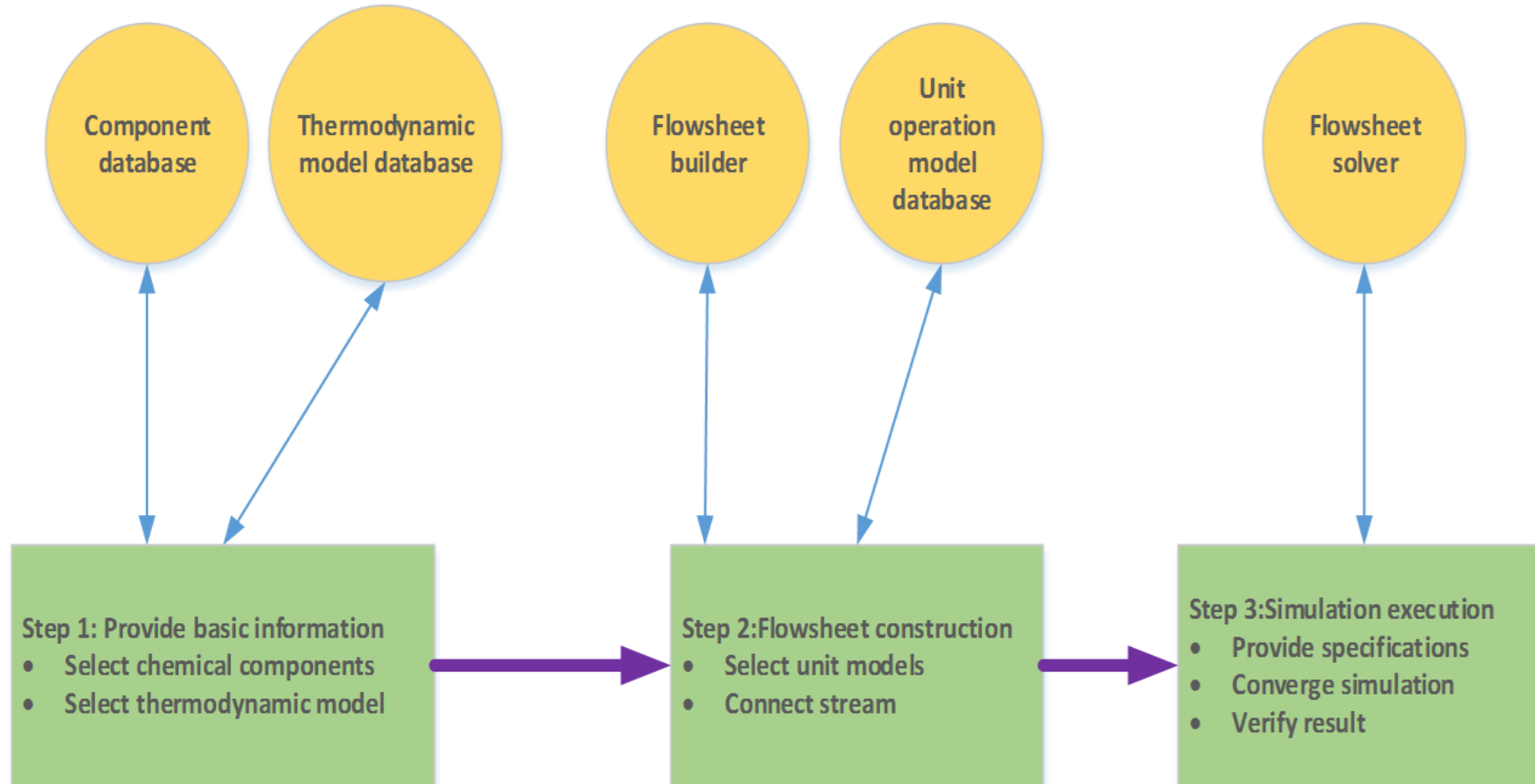
- In-built directionality from inlets to outlets make 'downstream' specifications (e.g. product specs) difficult
- Recycles may be (very) slow to converge, or fail to converge
- Poor handling of multiple or complex recycles

- Difficult to add new custom models; the user needs to code the solution method as well as model physics and chemistry
- Optimization capability is very limited
- Many other limitations for more complex problems.

■ Example of SM software

- Aspen Plus
- Aspen HYSYS
- Pro/II
- UniSim
- [COCO Simulator](#) (Freeware)
- [DWSIM](#) (Freeware)
- https://en.wikipedia.org/wiki/List_of_chemical_process_simulators

Basic Structure and Steps for Simulation Problem



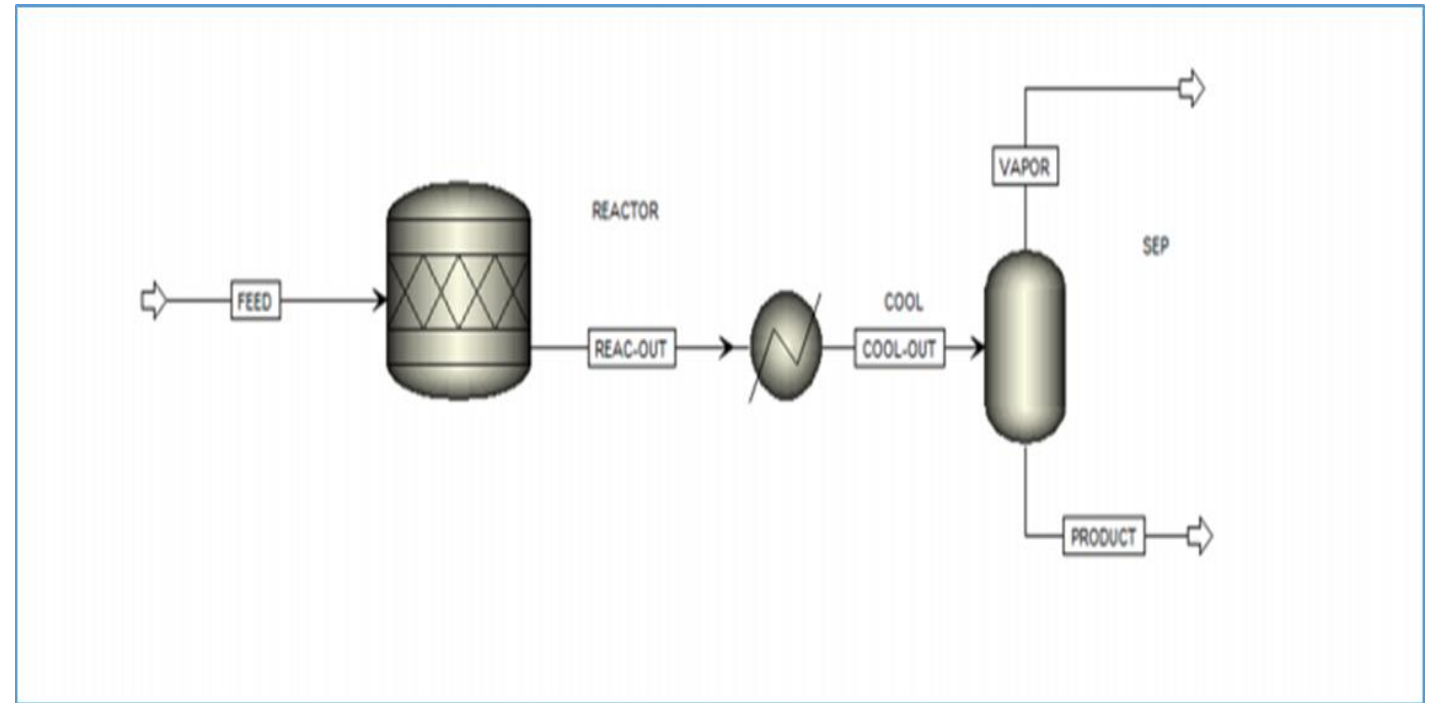
10 Good Habits for Process Simulation

- You build a simulation model to meet an objective
- Identify the system or process and draw an envelope around it
- Imagine what is going on physically
- Translate the physical model to a mathematical model
- Know your components
- Know the context of your feed streams
- Know your components bp
- Keep track of the units of measure in all calculations
- Always do a simple material and energy balance first
- Plot the phase envelope for important streams

1.2 Getting Started with Aspen Plus

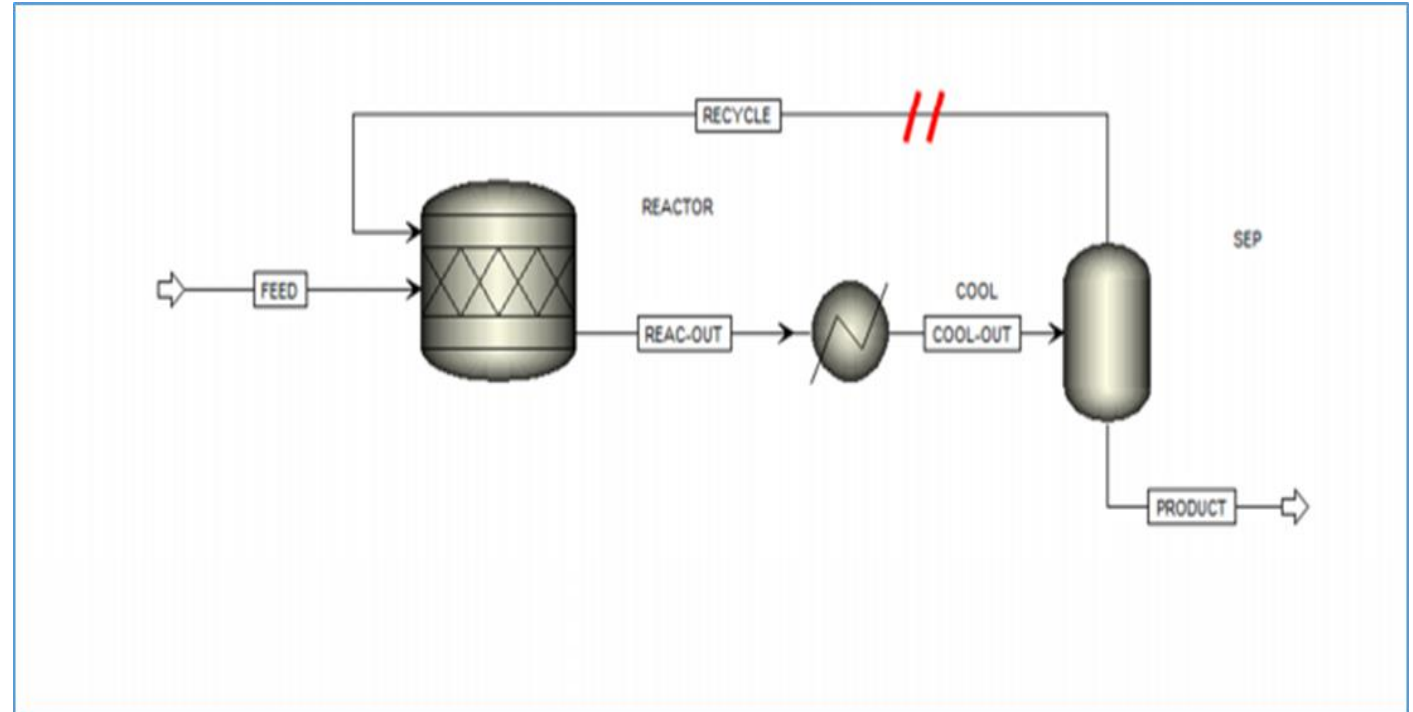
A Simple Simulation Problem

- What is the composition of the product stream?
- To answer this question, we need to solve the material and heat balances for each operation
- SM mode



A Simple Simulation Problem with Recycle

- What is the composition of the product stream?
- To answer this question, we need to solve the material and heat balances for each operation
- SM mode
- Give some estimates for the recycle stream and let convergence method close the balances

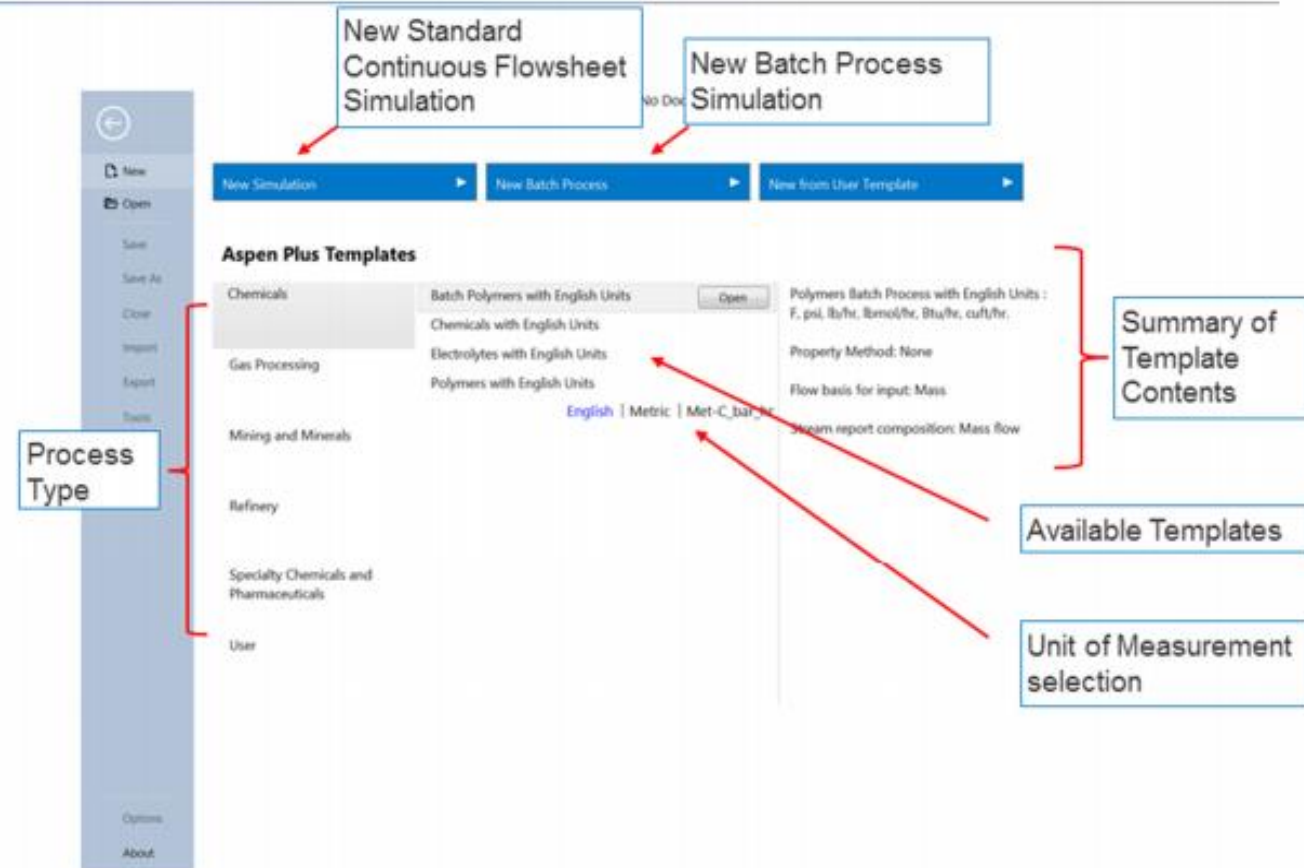


Starting Up Aspen Plus

- From the start menu go to...
 - Aspen Plus | Aspen Plus V12
 - For earlier version like V9, go to Aspen Plus | Aspen Plus V9
- Aspen Plus Start Page opens (next slide)
 - Start new simulation or..
 - Choose from previously open simulation
- Alternatively, just double click the simulation file in Windows folder.

Building a New Simulation

- Startup Aspen Plus
- Click on “New...” to create a new simulation
- Select the appropriate Template to begin the simulation



The screenshot shows the 'New Simulation' dialog box in Aspen Plus. The 'New Simulation' tab is selected, and the 'New Standard Continuous Flowsheet Simulation' button is highlighted. The 'New Batch Process Simulation' button is also visible. The 'Aspen Plus Templates' section is expanded, showing a list of templates under the 'Chemicals' category. The 'Batch Polymers with English Units' template is selected, and the 'Open' button is visible. The 'Summary of Template Contents' section is expanded, showing details for the selected template, including the property method (None), flow basis (Mass), and stream report composition (Mass flow). The 'Unit of Measurement selection' section is also visible, showing the 'English' unit system selected.

New Standard Continuous Flowsheet Simulation

New Batch Process Simulation

Process Type

Aspen Plus Templates

Summary of Template Contents

Available Templates

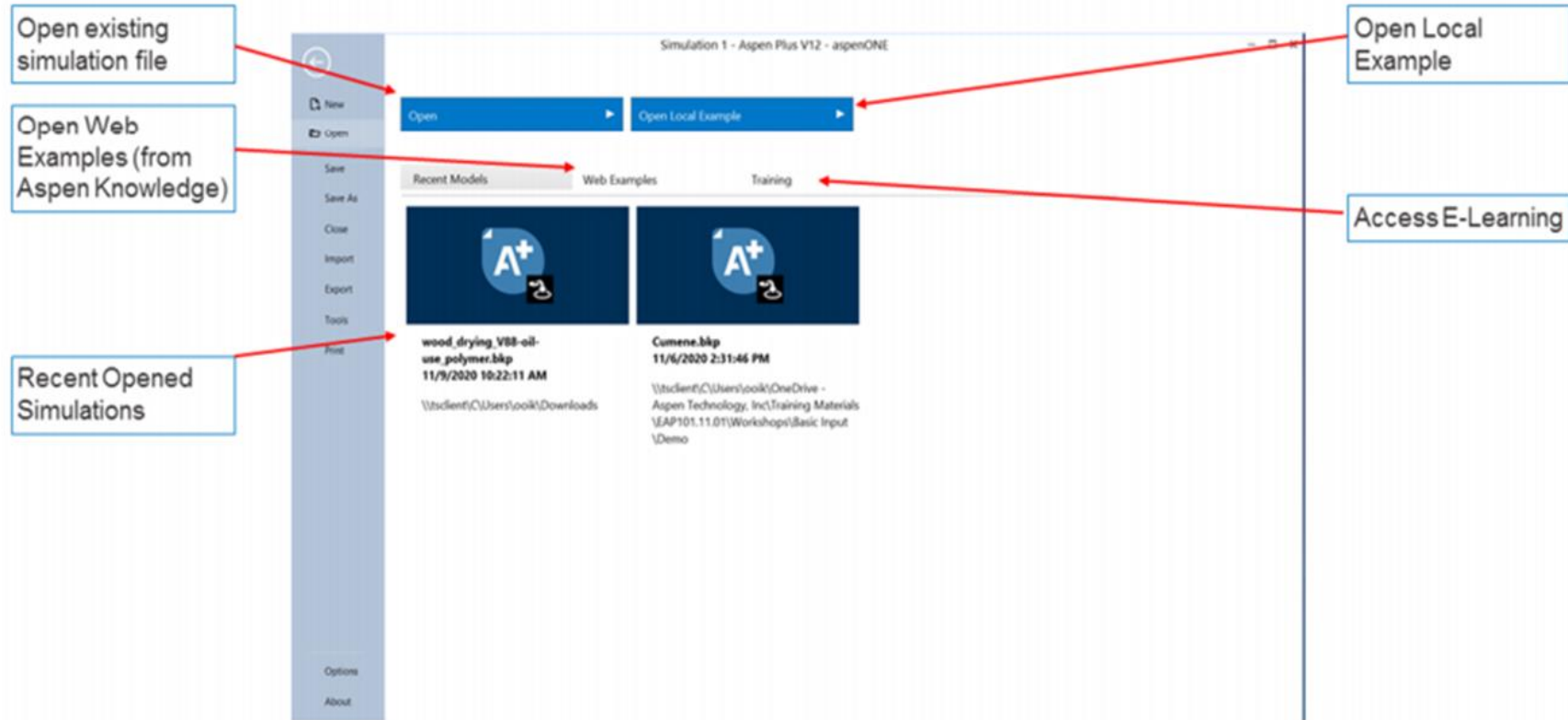
Unit of Measurement selection

Selecting a Template

- Template set defaults commonly used by specific industries for:
 - Units of measurement
 - Stream composition information and properties to report
 - Stream report format
 - Default setting for free water option
 - Property method
 - Other application specific defaults
- You can also customize the application template files

Open Existing or Example Simulation

V12



The screenshot shows the Aspen Plus V12 interface with the following annotations:

- Open existing simulation file:** Points to the **Open** button in the top toolbar.
- Open Web Examples (from Aspen Knowledge):** Points to the **Web Examples** tab in the Recent Models section.
- Recent Opened Simulations:** Points to the **Recent Models** tab in the Recent Models section.
- Open Local Example:** Points to the **Open Local Example** button in the top toolbar.
- Access E-Learning:** Points to the **Training** tab in the Recent Models section.

The Recent Models section displays two simulation files:

Simulation Name	Date/Time	Path
wood_drying_V88-oil-use_polymer.bkp	11/9/2020 10:22:11 AM	\\tsclient\C\Users\look\Downloads
Cumene.bkp	11/6/2020 2:31:46 PM	\\tsclient\C\Users\look\OneDrive - Aspen Technology, Inc.\Training Materials\EA101.11.01\Workshop\Basic Input Demo

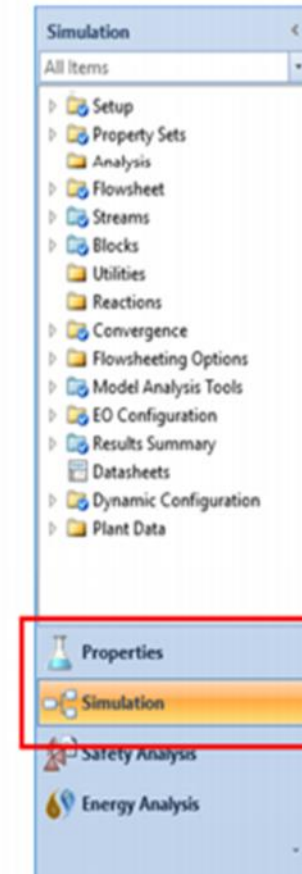
The User Interface

The screenshot displays the Aspen Plus V12 user interface, which is a software tool for process simulation. The interface is organized into several key areas:

- File Button:** Located at the top left, it provides access to file operations.
- Quick Access Toolbar:** A set of icons for frequently used functions like Save, Undo, and Redo.
- Ribbon Tabs:** These tabs (File, Home, Economics, Batch, Dynamics, Plant Data, Equation Oriented, View, Customize, Resources, Modify, Format) categorize the software's features.
- Contextual Ribbon Tabs:** These tabs appear based on the current selection or active analysis, such as 'Simulation' or 'Exchanges'.
- Min/Max Ribbon:** Controls for minimizing or maximizing the ribbon area.
- Display Help:** A button to access the software's help documentation.
- Ribbon:** The main area containing various tool icons for creating and editing process flowsheets.
- Activated Analysis:** A section showing the current analysis being performed, such as 'Energy Savings'.
- Navigation Pane:** A sidebar on the left that allows users to navigate through different project components like Setup, Property Sets, Analysis, Flowsheet, Streams, Blocks, Sensors, Utilities, Reactions, Convergence, Flowsheeting Options, Model Analysis Tools, 3D Configuration, Results Summary, Dynamic Configuration, and Plant Data.
- Environment:** A section for configuring the simulation environment, including 'Simulation', 'Safety Analysis', and 'Energy Analysis'.
- Status Bar:** Located at the bottom, it provides information about the current state of the simulation, such as 'Results Available' and 'Check Status'.
- Continuous Process Flowsheet:** The central workspace where the process flowsheet is built. It shows a sequence of units: a 'FEED' stream entering a 'REACTOR', followed by a 'REAC-OUT' stream, a 'COOL-OUT' stream, and finally a 'PRODUCT' stream. A 'RECYCLE' stream is also shown returning from the product to the reactor inlet.
- Model Palette Tabs:** These tabs (Mixers/Splitters, Separators, Exchanges, Columns, Reactors, Pressure Changers, Manipulators, Solids, Solids Separators, User Models, Batch Models) provide access to different types of process units.
- Model Palette:** A collection of icons representing various process units available for selection.
- Zoom Slider:** A control at the bottom right for adjusting the zoom level of the flowsheet.

Environments in Aspen Plus

- **Properties Environment** – all the steps needed to fully define the physical properties
 - Component selection and characterization
 - Select property methods and databases
 - Property estimation / structure drawing
 - Property analysis
 - Property data regression
- **Simulation Environment** – used to build and run models
 - Build and specify process flowsheets
 - Create and run interactive analysis (sensitivity / case studies etc.)
 - Fit models against process or lab data
 - Preliminary equipment design, sizing and rating
 - Relative cost estimation (economic evaluation)



Building an Aspen Plus Simulation

- **Properties Environment:**

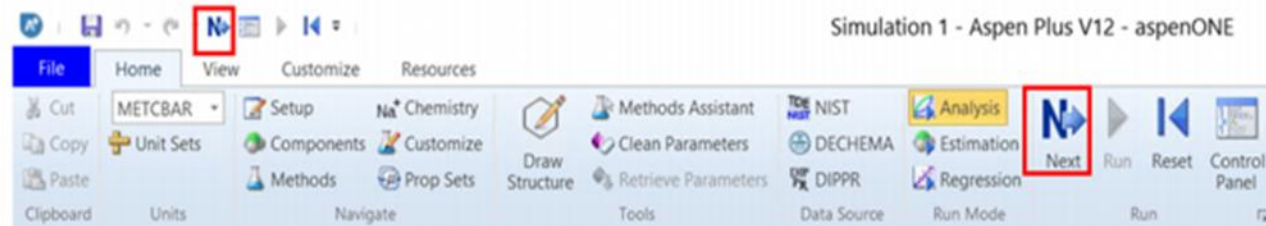
- Specify the chemical components in the process
 - Aspen Plus databanks or user defined
- Specify the thermodynamic models to represent physical properties components and mixtures in the process
 - Built-in models or custom models

- **Simulation Environment:**

- Define the process flowsheet:
 - Unit operations and stream flow connections
 - Graphically build flowsheet with models from the Model Palette
- Specify feed stream component flows and conditions
- Specify operating conditions and other parameters for unit operation models
- Optional: Specify process utilities and cost factors if performing energy and cost analysis

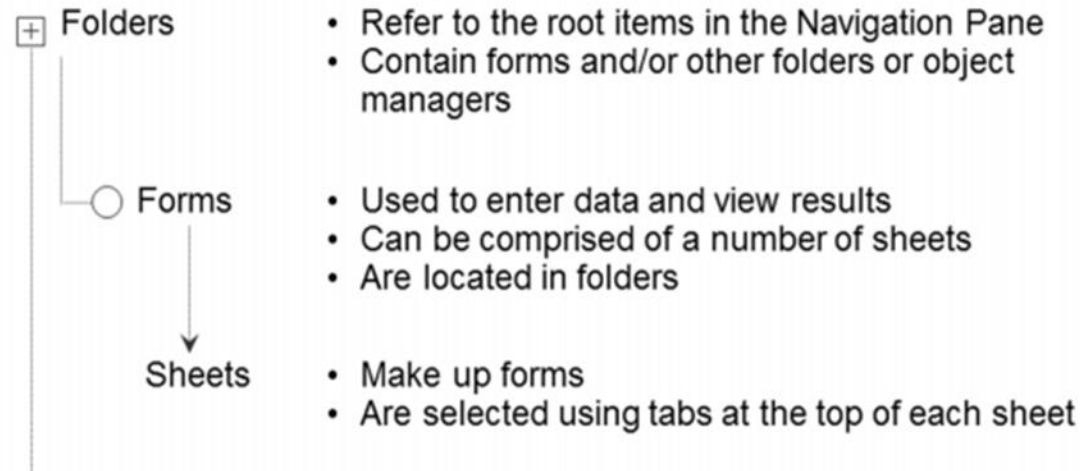
Aspen Plus Interface

- Graphical User Interface
 - Ribbons to launch actions
 - Model Palette to build the process flowsheet
- Navigation Pane
 - Used to navigate the folders, forms, and sheets
 - Contains Object Managers, which allow for manipulation of discrete objects of information
- Next Button: Checks if the current form is complete and moves to the next form which requires input

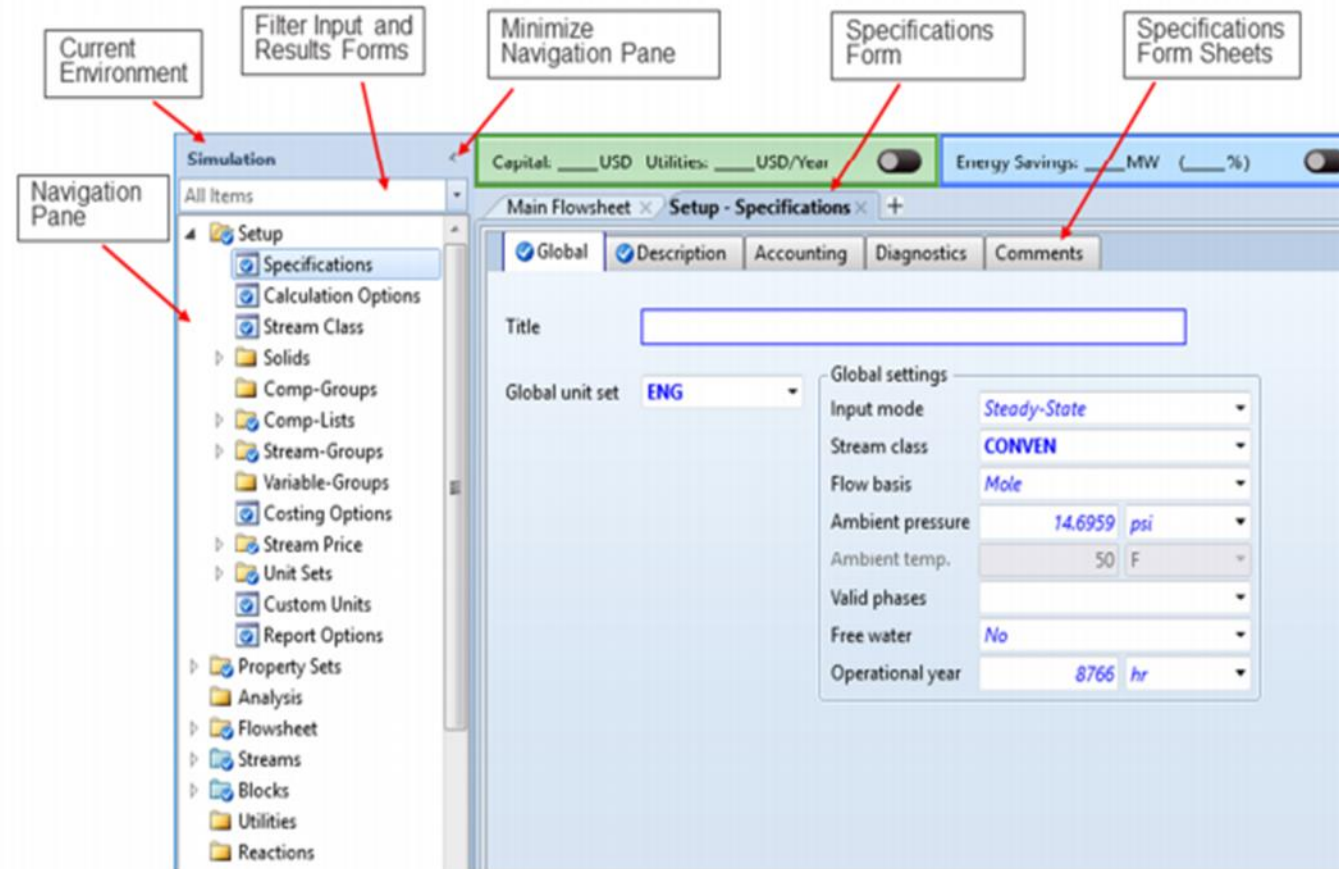


Navigation Pane (1)

- The Navigation Pane is a hierarchical tree structure view of the available simulation input, results, and objects that have been defined; it is used to navigate the folders, forms, and sheets



Navigation Pane (2)



Status Indicators

This Symbol	On a(n)	Means
 	Input form or sheet	Required input complete, or visited and no data required *
 	Input form or sheet	Required input incomplete
	Input form	No data entered
	Mixed form	Input and results
	Results form	No results present (calculations have not been run)
	Results form	Results available without Errors or Warnings (OK) **
 	Results form or flowsheet	Results available with Warnings **
 	Results form or flowsheet	Results available with Errors **
	Results form	Results inconsistent with current input (input changed)
 	Input folder	No data entered
 	Input folder	Required input incomplete
 	Input folder	Required input complete, or visited and no data required *
	Results folder	No results present
 	Results folder	Results available – OK **
 	Results folder	Results available with Warnings **
 	Results folder	Results available with Errors **
 	Results folder	Results inconsistent with current input (input changed)
 	Folder or form	Object deactivated

Help

- Help Topics

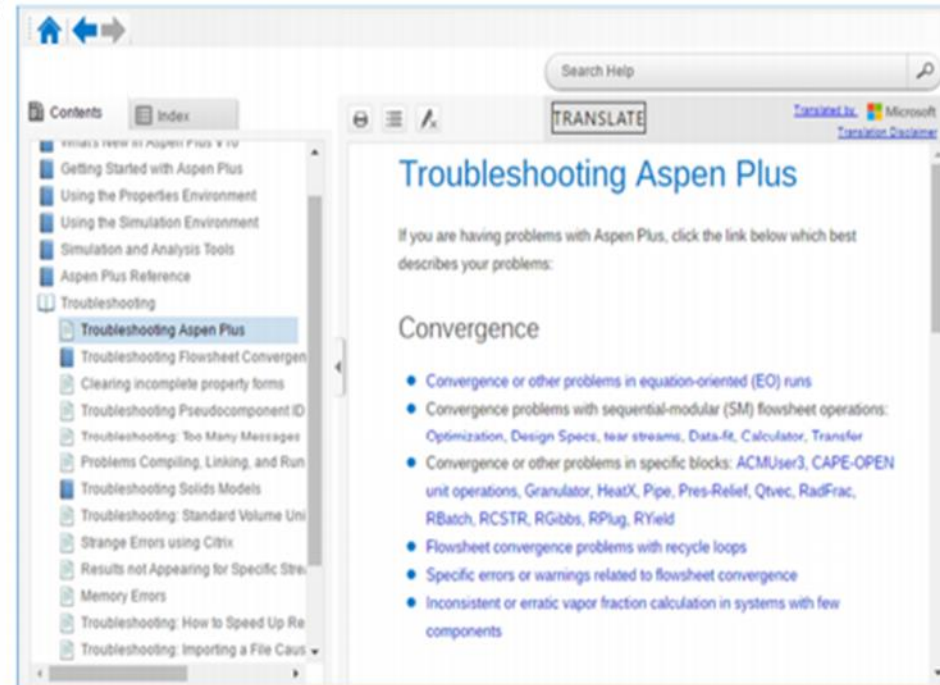
- Select Help Topics from the Help button on the ribbon  to launch online help:
 - Contents: Browse through the documentation, including User Guides and Reference Manuals
 - Index: Search for help on a specific topic using the index entries
 - Search: Search for a help on a topic that includes any word or words

- F1 Help

- With the cursor in the desired field, press the <F1> function key to bring up help for field and/or sheet

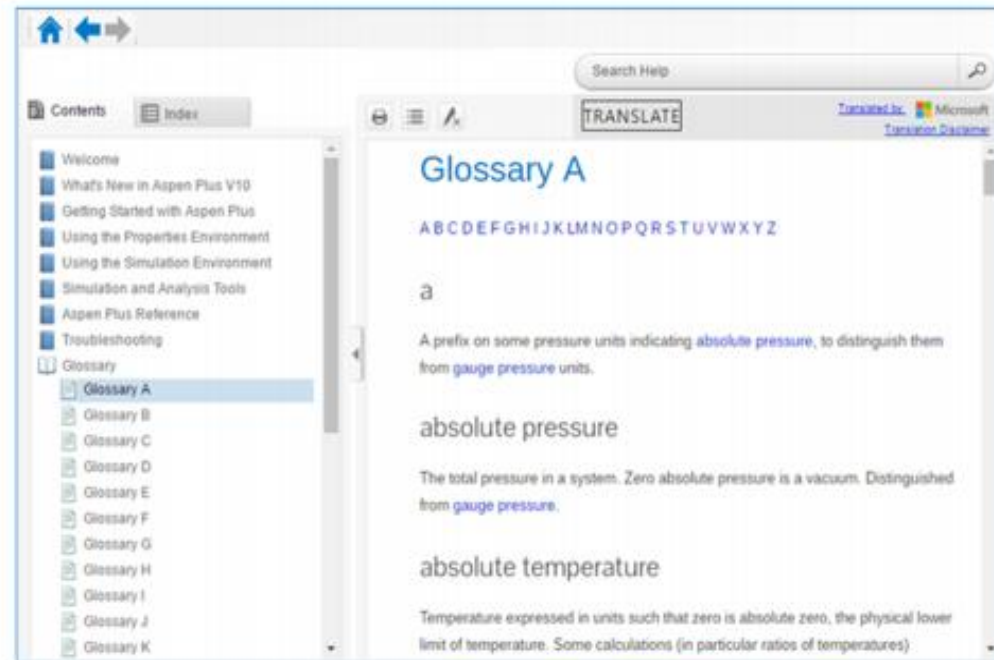
Troubleshooting Help

- Troubleshooting using the online help
 - Based on specific error and warning messages
 - Based on a description of the problem
 - Using diagnostic messages and results of failed runs



Glossary Help

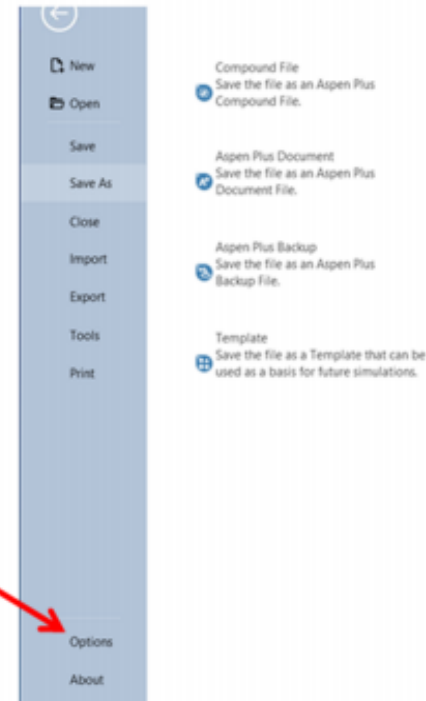
- Glossary of terms used in Aspen Plus documentation
- Cross-linked to other terms



Saving an Aspen Plus Simulation

- To save a file:
 - Select Save As from the File menu
 - Choose an appropriate Save As Type
 - Choose a File name

Default file save type and other preferences can be set from the **File | Options** dialog box



File Formats in Aspen Plus Simulations

File Type	Extension	Format	Description
Compound	*.apwz	Binary	Compressed file which contains the model (the BKP or APW file) and external files referenced by the model. You can add additional files such as supporting documentation to the APWZ file.
Document	*.apw	Binary	File containing simulation input, results and intermediate convergence information
Backup	*.bkp	ASCII	Archive file containing simulation input and results
Template	*.apt	ASCII	Template containing default inputs

Compound Files (.apwz)

Problem:

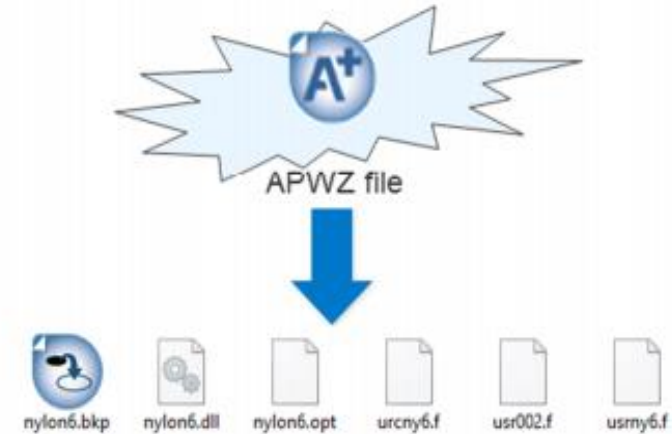
It can be difficult to manage all the files associated with a simulation

- A single Aspen Plus case may involve dozens of files:
 - Backup file (.bkp)
 - Embedded backup file (.apmbd)
 - FORTRAN files (.dll, .dlopt, .obj, .F)
 - Exchanger design and rating files (.edr, etc)
 - Economic evaluation files (.izp, .szp, etc.)
 - etc.

Solution:

Embed all the dependent files into a single "Compound File"

- Aspen automatically identifies the files to be included in the compound case file
- Manual over-ride to allow user to add or remove content

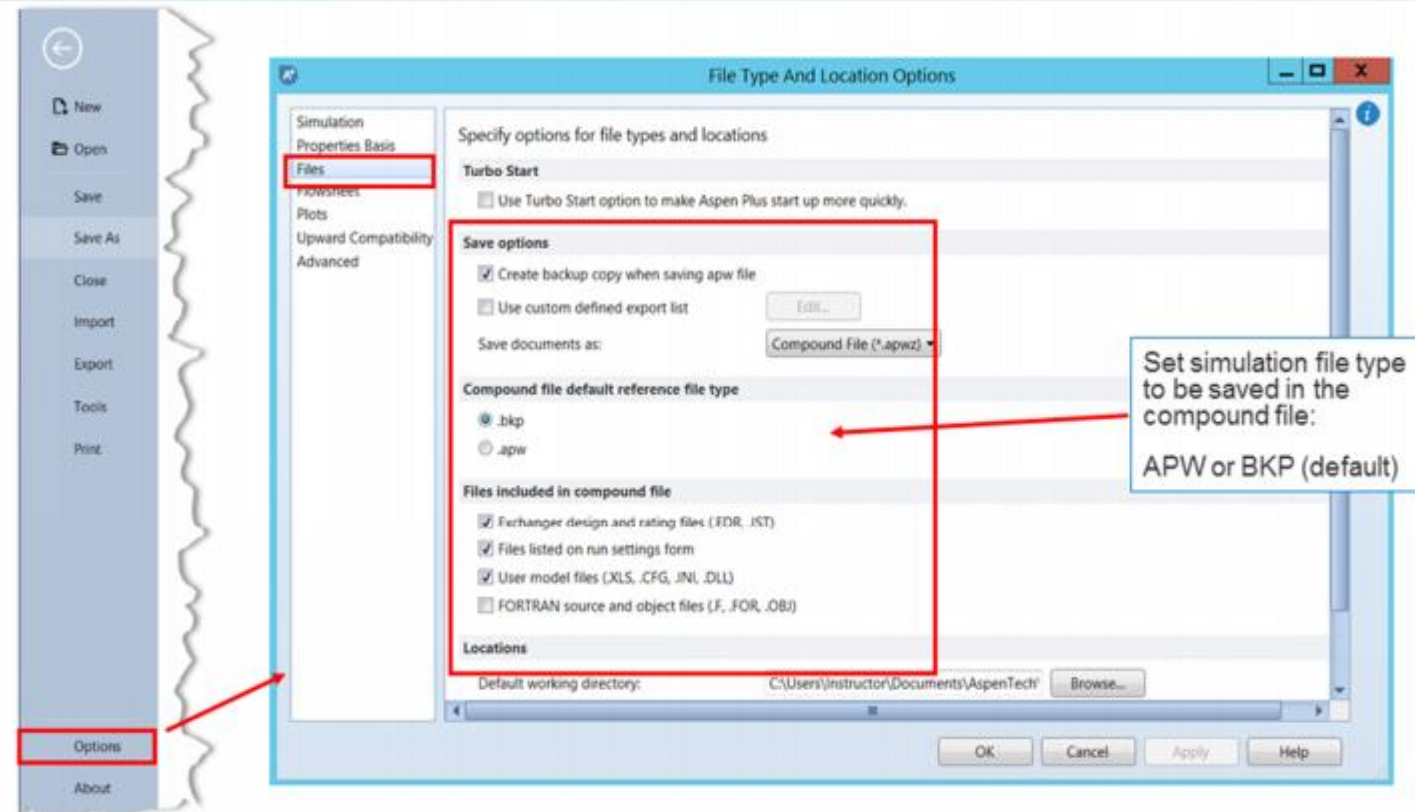


Benefits:

Easy file management and deployment

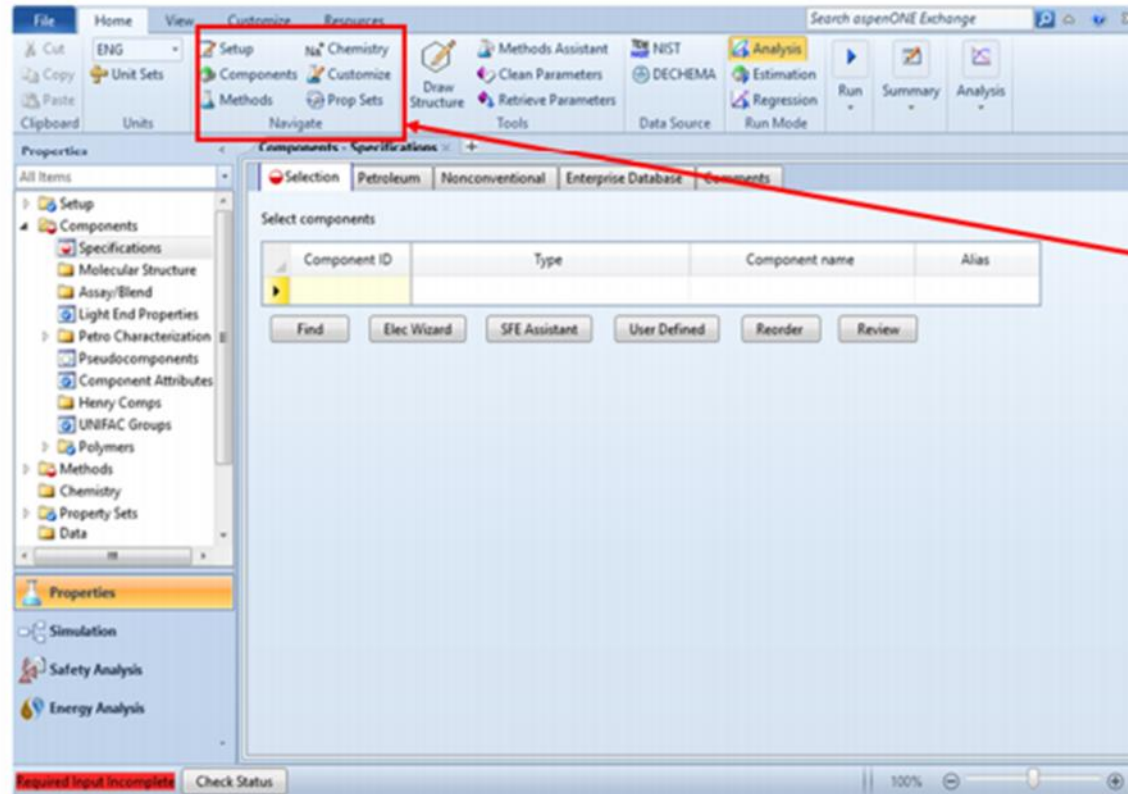
- E-mail
- File shares
- Document management systems such as SharePoint, Documentum

Default File Save Options and Preferences



1.3 Thermodynamic in Aspen Plus

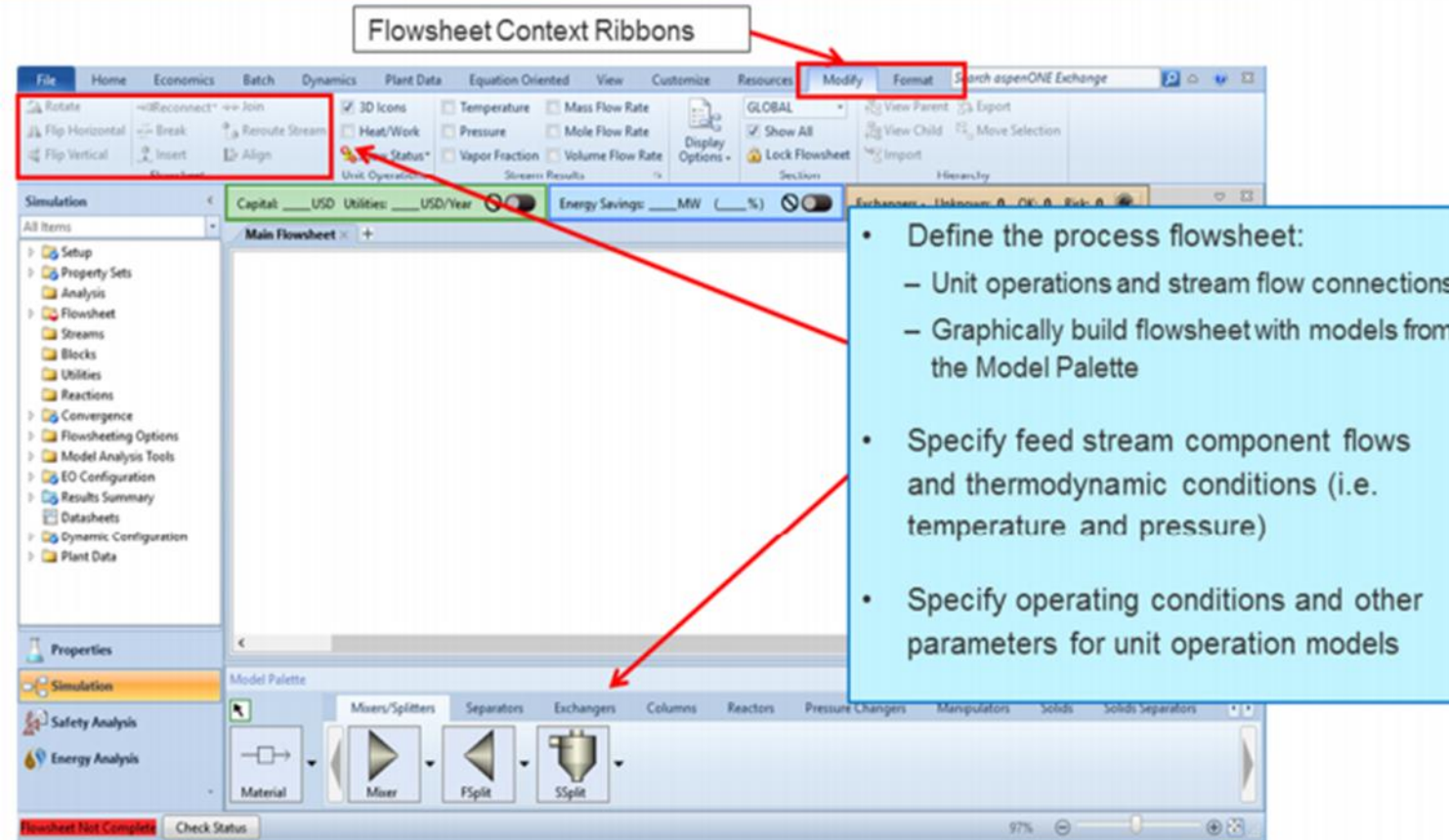
Properties Environment



- Specify the chemical components in the process
 - Aspen Plus databanks or user defined
- Specify the thermodynamic models to represent physical properties components and mixtures in the process
 - Built-in models or custom models

Simulation Environment

Flowsheet Context Ribbons



- Define the process flowsheet:
 - Unit operations and stream flow connections
 - Graphically build flowsheet with models from the Model Palette
- Specify feed stream component flows and thermodynamic conditions (i.e. temperature and pressure)
- Specify operating conditions and other parameters for unit operation models

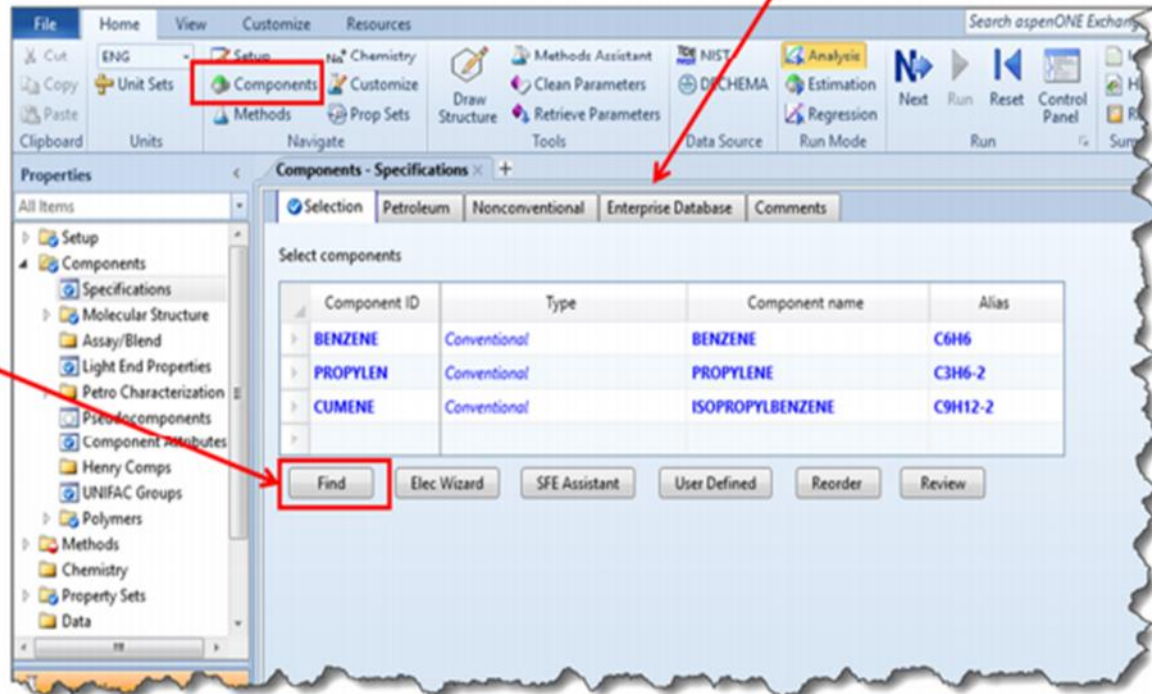
Components

- Start in the Properties Environment
- Use the Components | Specifications | Selection sheet to specify all the components required for the simulation
- If available, physical property parameters for each component are retrieved from databanks
- Pure component databanks contain parameters such as molecular weight, critical properties, etc; the databank search order is specified on the Enterprise Database sheet.
- The Find button can be used to search for components
- The Elec Wizard can be used to set up an electrolyte simulation
- The SPE Assistant can be used to set up chemistry for solid-fluid equilibrium
- The User Defined button to create a non-databank component

Components Specifications Form

Property Databanks

Search Property Databank



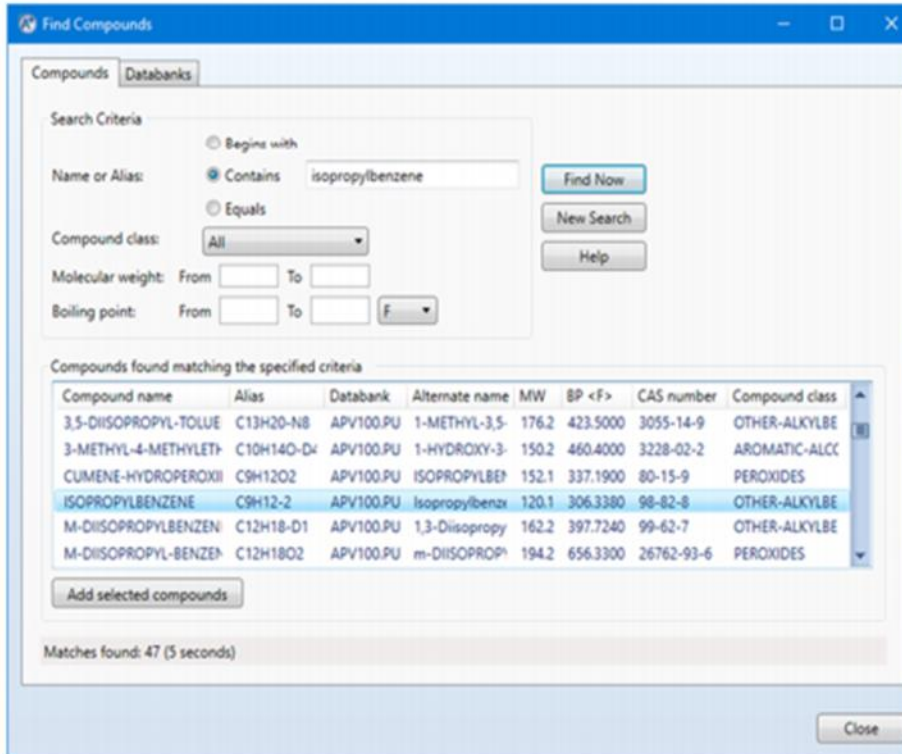
Find Elec Wizard SFE Assistant User Defined Reorder Review

Component ID	Type	Component name	Alias
BENZENE	Conventional	BENZENE	C6H6
PROPYLEN	Conventional	PROPYLENE	C3H6-2
CUMENE	Conventional	ISOPROPYLBENZENE	C9H12-2

Entering Components

- The component ID is used to identify the component in simulation inputs and results.
- Each Component ID can be associated with databank component as either:
 - Alias: Chemical formula of component (e.g., C₆H₆)
 - ❖ A suffix is added to formulas when there are isomers, e.g., C₂H₆O-2
 - Component Name: Full-name of component (e.g., BENZENE)
- Databank components can be searched for using the Find button.

Find Using Aspen Properties Enterprise Database



Find Compounds

Compounds Databanks

Search Criteria

☐ Begin with
☒ Contains isopropylbenzene
☐ Equals

Compound class: All

Molecular weight: From To

Boiling point: From To F

Find Now

New Search

Help

Compounds found matching the specified criteria

Compound name	Alias	Databank	Alternate name	MW	BP <F>	CAS number	Compound class
3,5-DIISOPROPYL-TOLUE	C13H20-NB	APV100.PU	1-METHYL-3,5-	176.2	423.5000	3055-14-9	OTHER-ALKYLBE
3-METHYL-4-METHYLETH	C10H14O-D4	APV100.PU	1-HYDROXY-3-	150.2	460.4000	3228-02-2	AROMATIC-ALCC
CUMENE-HYDROPEROXII	C9H12O2	APV100.PU	ISOPROPYLBE ¹	152.1	337.1900	80-15-9	PEROXIDES
ISOPROPYLBENZENE	C9H12-2	APV100.PU	Isopropylbenz ¹	120.1	306.3380	98-82-8	OTHER-ALKYLBE
M-DIISOPROPYLBENZEN	C12H18-D1	APV100.PU	1,3-Diisopropy	162.2	397.7240	99-62-7	OTHER-ALKYLBE
M-DIISOPROPYL-BENZE ¹	C12H18O2	APV100.PU	m-DIISOPROP ¹	194.2	656.3300	26762-93-6	PEROXIDES

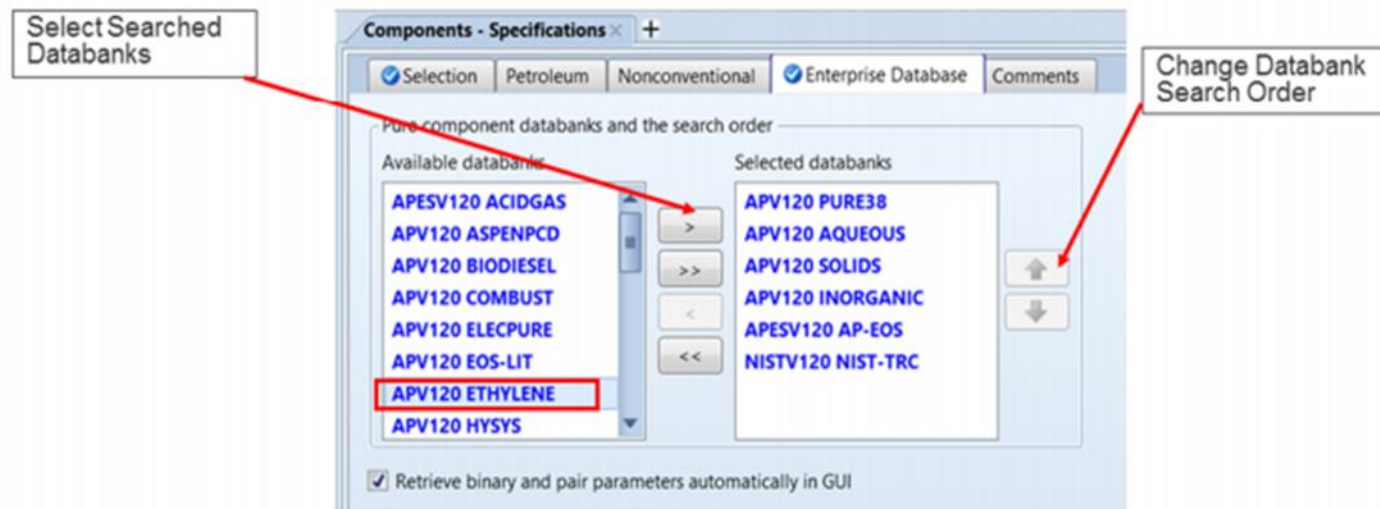
Add selected compounds

Matches found: 47 (5 seconds)

Close

- Search using component name, alias, component class, molecular weight, boiling point, or CAS number
- Find** performs an AND search when more than one criterion is specified
- All components containing specified items will be listed

Pure Component Databanks



- Parameters missing from the first selected databank will be searched for in subsequent selected databanks

List of Pure Component Databanks (1)

Databank	Contents	Uses
PURE38	Data from the Design Institute for Physical Property Data (DIPPR) and AspenTech	Primary component databank in Aspen Plus
AQUEOUS	Pure component parameters for ionic and molecular species in aqueous solution	Simulations containing electrolytes
SOLIDS	Pure component parameters for strong electrolytes, salts, and other solids	Simulations containing electrolytes and solids
INORGANIC	Thermochemical properties for inorganic components in vapor, liquid, and solid states	Solids, electrolytes, and metallurgy applications
AP-EOS	Pure component and binary interaction parameters used in the CPA equation of state property method	Mixtures containing hydrocarbons and polar/associating chemicals
NIST-TRC	Data from the National Institute of Standards and Technology (NIST) Thermal Research Center (TRC)	Vast range of components

- Older PURExx databanks are included for upward compatibility

List of Pure Component Databanks (2)

Databank	Contents	Uses
ETHYLENE	Pure component and binary interaction parameters for components typically found in ethylene processes for use with the SRK property method	Ethylene process simulation
COMBUST	Pure component parameters for components typically found in combustion products, including free radicals	Simulations containing combustion products
POLYMER	Chains of repeating monomer units	Polymers processes
SEGMENT	Individual monomer groups or segments	Polymers processes
INITIATOR	Property parameters and thermal decomposition reaction rate parameters for polymer initiator species.	Polymers processes
ELECPURE	Pure component parameters for some components commonly found in amine processes.	Amine processes
FACTPCD	FACT species (components referenced in a specific pure or solution phase for use only with the Aspen/FACT/Chemapp interface in Aspen Plus)	Pyrometallurgical processes

List of Pure Component Databanks (3)

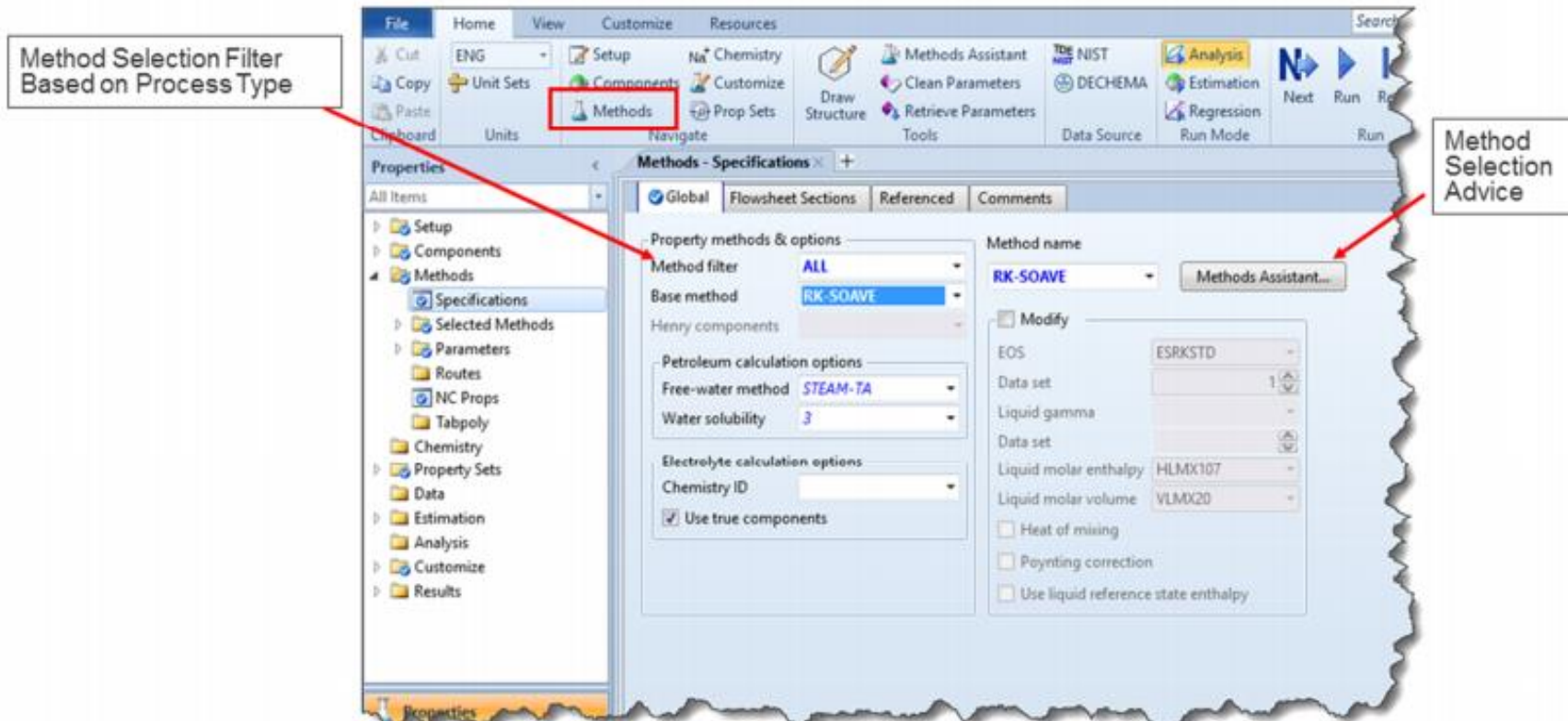
Databank	Contents	Uses
PPDS	Physical Property Data System databank of the National Engineering Laboratory	Link to PPDS databank
NRTL-SAC	Pure component parameter XYZE containing segment representations of common solvents	Calculations using NRTL-SAC property method
HYSYS	Pure component and binary parameters needed by Aspen HYSYS property methods	Models using Aspen HYSYS property methods
BIODIESEL	Pure component parameters for components typically found in biodiesel production processes	Biodiesel processes

- HYSYS databank used to support Aspen Properties in HYSYS

Property Methods

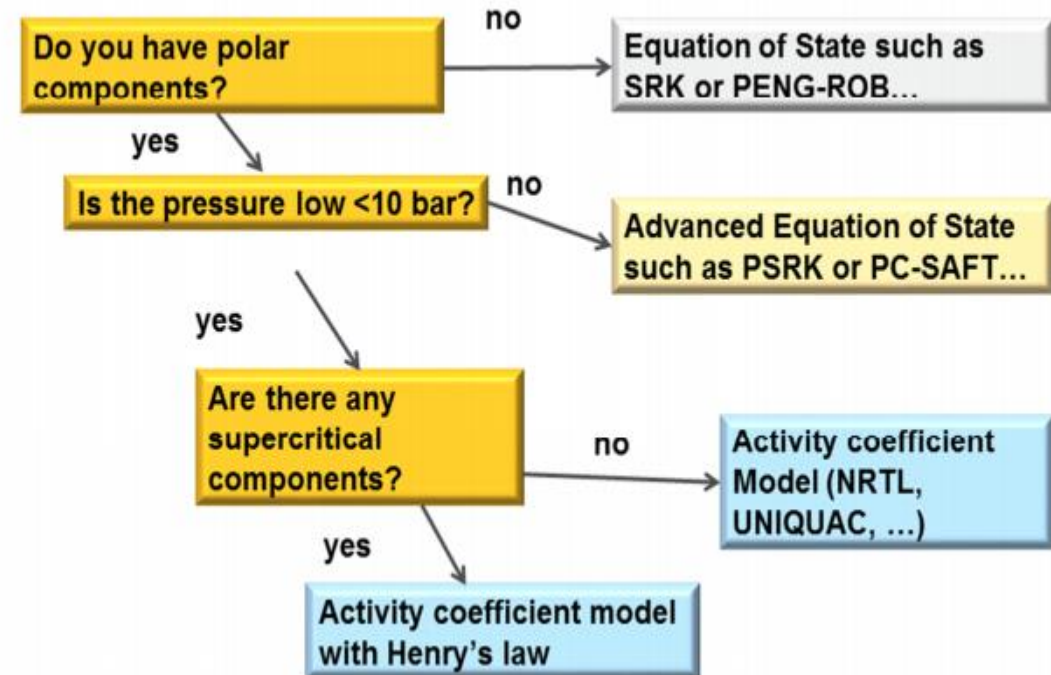
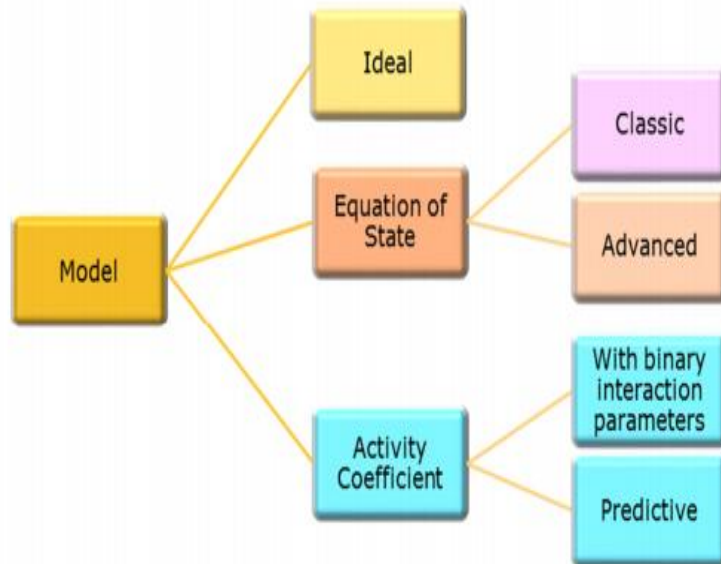
- Use the Methods | Specifications form to specify the physical property methods to be used in the simulation.
- Property methods are a collection of models and methods used to describe pure component and mixture behavior, and to calculate physical properties. Example of physical property is K-values for distillation unit operation.
- Having the correct physical properties is critical for obtaining reliable simulation results.
- Property Methods containing commonly used thermodynamic models are provided in Aspen Plus.

Methods Specifications Form



Approaches to Representing Physical Properties of Components

- Choice of model type depends on degree of non-ideal behavior, operating conditions, and availability of parameters.



Ideal vs Non-Ideal Behavior

- What do we mean by ideal behavior?
 - Ideal Gas Law
 - Raoult's Law
- Which systems behave as ideal?
 - Non-polar components of similar size and shape
- What control degree of non-ideality?
 - Molecular interactions, e.g., polarity, size and shape of the molecules.
- How can we study the degree of non-ideality of a system?
 - Property analysis plots (e.g., TXY & XY)

Features of EOS and Activity Models

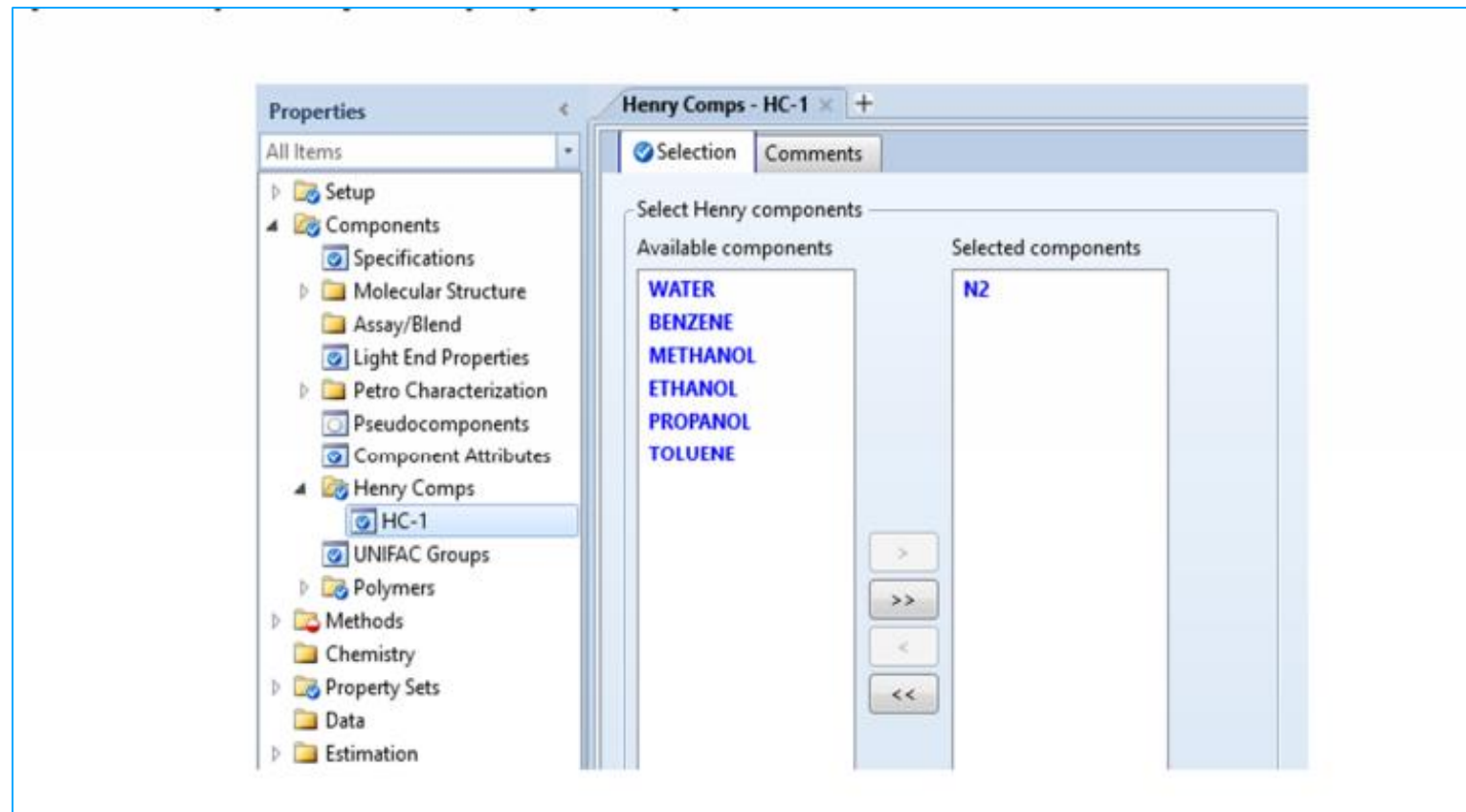
Equation of State Models	Activity Coefficient Models
Good for vapor phase modeling and liquids of low polarity	Good for liquid phase modeling only
Typically limited in ability to represent non-ideal liquids	Can represent highly non-ideal liquids
Fewer binary parameters required	Many binary parameters required
Parameters extrapolated reasonably with temperature	Binary parameters are highly temperature dependent
Consistent in critical region	Inconsistent in critical region
Examples: - PENG-ROB - RK-SOAVE - PC-SAFT - PSRK	Examples: - NRTL - UNIQUAC - WILSON - UNIFAC

Henry's Law

- Henry's Law is used to determine the amount of a supercritical component or light gas in the liquid phase.
 - Typically used with Ideal and Activity Coefficient models
 - Activity Coefficient model: $y_i P = x_i y_i P^{L_i}$ P^{L_i} = vapor pressure
 - Henry model: $y_i P = x_i H_i$ where $H_i = f(T)$
 - H is calculated from temperature-dependent Henry's constants for each solute-solvent pair
 - Declare any supercritical components or light gases (CO₂, N₂, etc) as Henry's components in the Properties Environment Inside Components | Henry Comps folder.

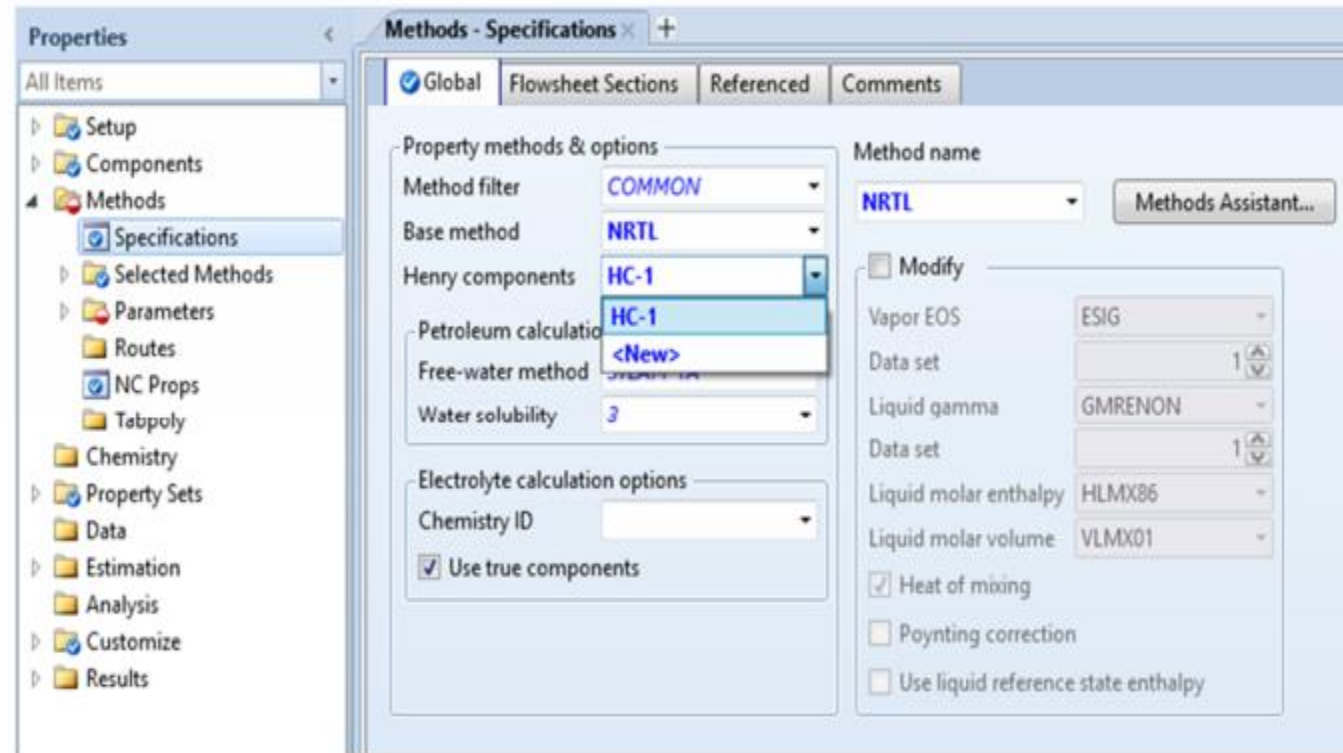
Henry's Law

- Declare any supercritical components or light gases (CO₂, N₂, etc) as Henry's components on the Components | Henry Comps | HC-X | Selection sheet.



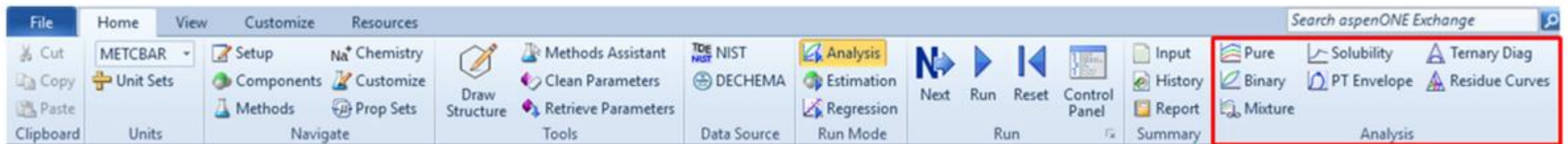
Henry's Law

- Then, select the Henry's components ID from the Henry Components dropdown list on the Methods | Specifications | Global sheet



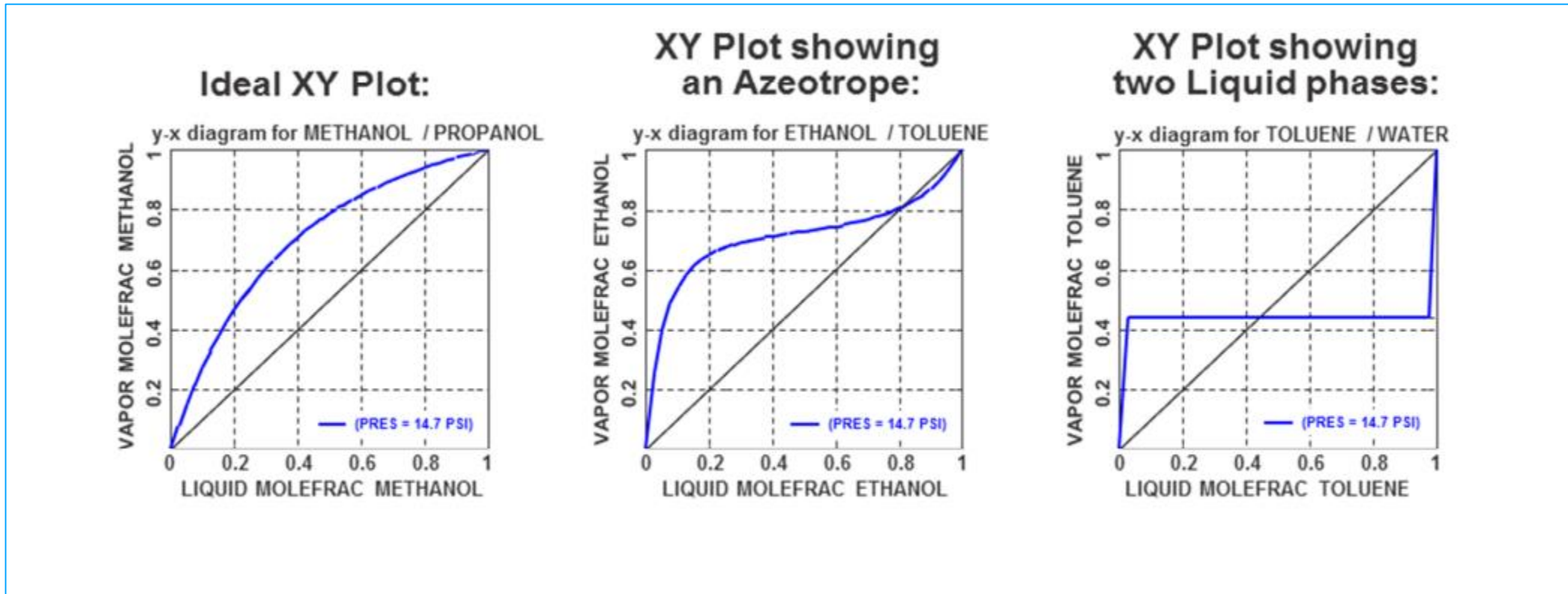
Property Analysis

- Used to generate simple property diagrams to validate physical property models and data.
- Diagram types:
 - Pure component, i.e., vapor pressure vs temperature
 - Binary, i.e., TXY, TXXY, PXY, and PXXY.
 - Mixture
 - Solubility
 - Ternary residue maps
 - Ternary phase diagrams
 - PT Envelope
- Property analysis input and results are saved as a form in the Analysis folder for later use



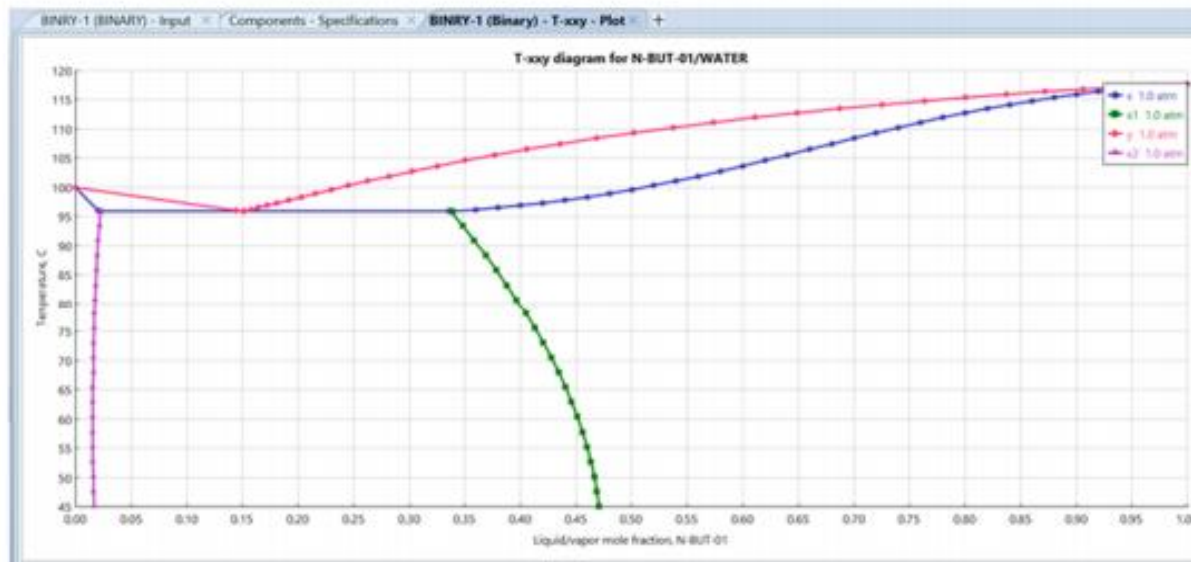
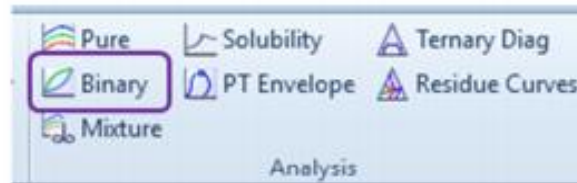
Property Analysis Plots

- Predicting non-ideal behavior:
- The default valid phases for binary analysis is vapor-liquid-liquid



VLLE Phase Diagrams

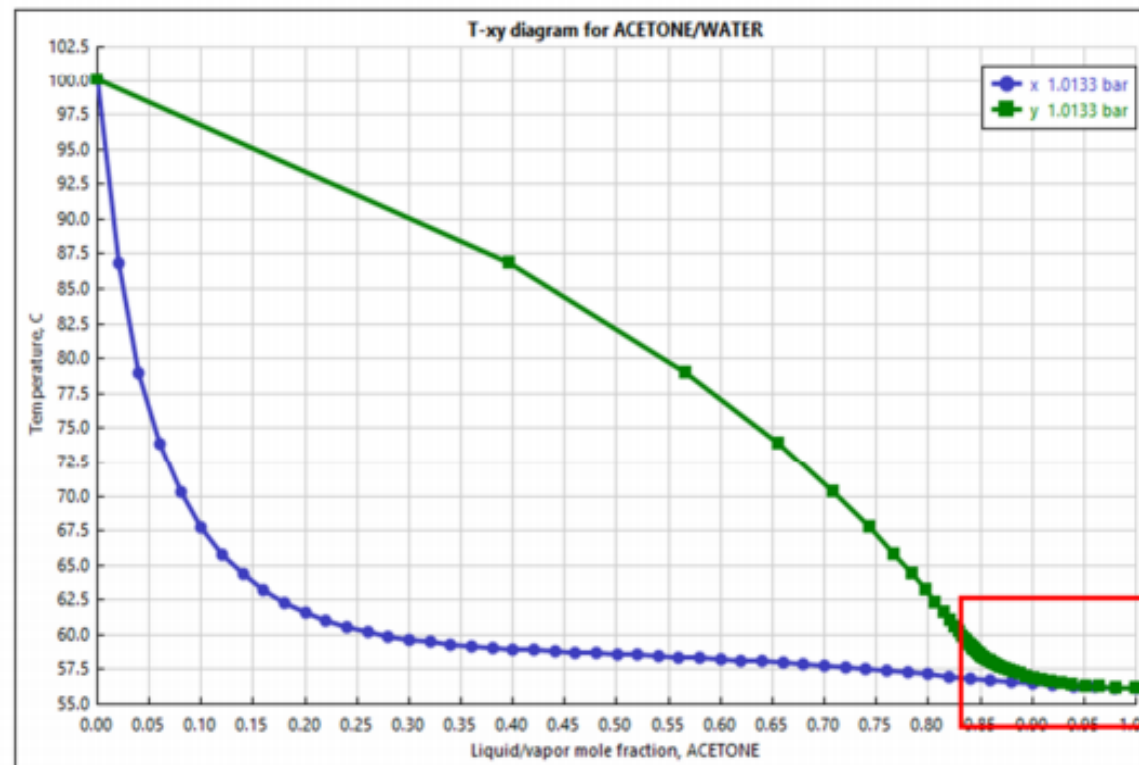
N-butanol/Water System
 (Using LLE Parameters)



Display VLE and LLE on one phase diagram

Why Does Acetone/Water Require Activity Coefficient Approach?

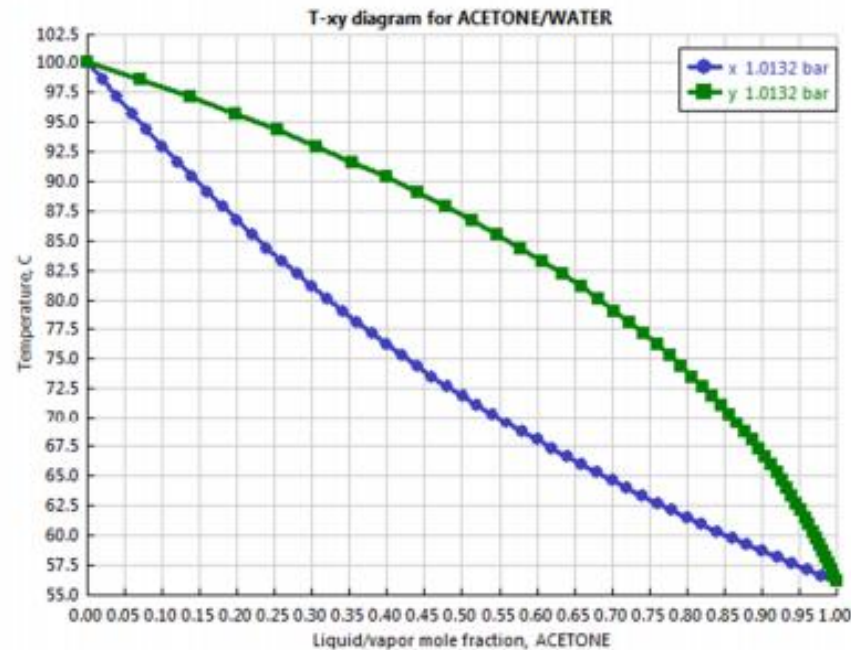
- Txy diagram of Acetone/Water at 1 atm using NRTL property method



Pinch point at high Acetone concentrations requires many stages to reach purity

Why Does Acetone/Water Require Activity Coefficient Approach?

T-xy Diagram of Acetone/Water at 1 atm using Ideal property method



T-xy Diagram of Acetone/Water at 1 atm using Equation of State (Peng-Robinson) property method



Property Method Selection Assistant

Property Method Selection Assistant

- Select **Methods Assistant** from the **Home** ribbon **Tools** group



- The assistant will help you select the most appropriate property method by guiding you through a number of questions
- Search by component type or process type
- At the end, the help topics for the recommended property methods are linked
- A report also is available that can be saved or printed

Recommended Property Methods for Different Applications

Gas Processing

Application

Hydrocarbon separations
Demethanizer
C3-splitter

Cryogenic gas processing
Air separation

Gas dehydration with glycols

Acid gas absorption with
Methanol (RECTISOL)
NMP (PURISOL)

Acid gas absorption with
Water
Ammonia
Amines
Amines + methanol
(AMISOL)
Caustic
Lime
Hot carbonate

Claus process

Recommended Property Methods

PR-BM, RKS-BM, PENG-ROB, RK-SOAVE

PR-BM, RKS-BM, PENG-ROB, RK-SOAVE

PRWS, RKSWS, PRMHV2, RKSMHV2, PSRK, SR-POLAR

PRWS, RKSWS, PRMHV2, RKSMHV2, PSRK, SR-POLAR

ELECNRTL

PRWS, RKSWS, PRMHV2, RKSMHV2, PSRK, SR-POLAR

Petrochemicals

Application	Recommended Property Methods
Ethylene plant Primary fractionator	CHAO-SEA, GRAYSON
Light hydrocarbons Separation train Quench tower	PENG-ROB, RK-SOAVE
Aromatics BTX extraction	WILSON, NRTL, UNIQUAC and their variances
Substituted hydrocarbons VCM plant Acrylonitrile plant	PENG-ROB, RK-SOAVE
Ether production MTBE, ETBE, TAME	WILSON, NRTL, UNIQUAC and their variances
Ethylbenzene and styrene plants	PENG-ROB, RK-SOAVE -or- WILSON, NRTL, UNIQUAC and their variances
Terephthalic acid	WILSON, NRTL, UNIQUAC and their variances (with dimerization in acetic acid section)

Chemicals

Application

Azeotropic separations
Alcohol separation

Carboxylic acids
Acetic acid plant

Phenol plant

Liquid phase reactions
Esterification

Ammonia plant

Fluorochemicals

Inorganic Chemicals
Caustic
Acids
Phosphoric acid
Sulphuric acid
Nitric acid
Hydrochloric acid

Hydrofluoric acid

Recommended Property Methods

WILSON, NRTL, UNIQUAC and their variances

WILS-HOC, NRTL-HOC, UNIQ-HOC

WILSON, NRTL, UNIQUAC and their variances

WILSON, NRTL, UNIQUAC and their variances

PENG-ROB, RK-SOAVE

WILS-HF

ELECNRTL

ENRTL-HF

Simulation Environment Interface

You can enter the simulation environment from here

Status indicator

Model palette enables you to select **material streams** and **unit operations (blocks)** to be added to your flowsheet

The screenshot displays the Aspen Plus V12 - aspenONE Simulation 1 interface. The top menu bar includes File, Home, Economics, Batch, Dynamics, Plant Data, Equation Oriented, View, Customize, Resources, Modify, and Format. The left sidebar shows a tree view of simulation components, including Setup, Property Sets, Analysis, Flowsheet, Streams, Blocks, Sensors, Utilities, Reactions, Convergence, Flowsheeting Options, Model Analysis Tools, EO Configuration, Results Summary, Datasheets, Dynamic Configuration, and Plant Data. The central workspace is titled 'Main Flowsheet' and contains a 'Model Palette' window. The 'Model Palette' window has tabs for Mixers/Splitters, Separators, Exchangers, Columns, Reactors, and Pressure Changers. The 'Mixers/Splitters' tab is active, showing icons for MATERIAL, Mixer, FSplit, and SSplit. Red arrows point from text annotations to specific parts of the interface: one to the 'Simulation' button in the left sidebar, one to the 'Status indicator' in the bottom bar, and three to the 'Model Palette' window.

Simulation Environment Interface

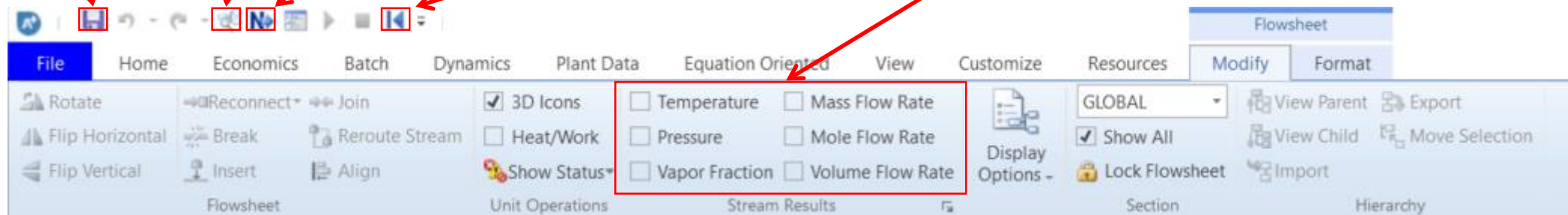
Save button
(best to save as
backup file)

Flowsheet
button (useful
when multiple
tabs are opened
or main
flowsheet
cannot be
found)

Next button [F4]
(takes you to
next minimum
input required to
complete
flowsheet)

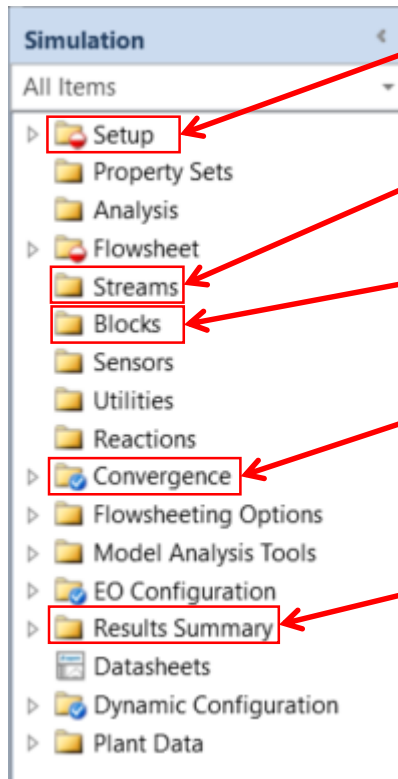
Reset button
(purges all
temporary or
calculated data
but not input by
user)

You may check
these boxes in
order to display
the
corresponding
stream
information on
the flowsheet



The top ribbon of the simulation environment is similar to most Microsoft Office products

Simulation Environment Interface



Specifications to be applied to entire flowsheet can be accessed here

Streams will be updated to this folder as you add them to the flowsheet

Blocks will be updated to this folder as you add them to the flowsheet

Troubleshooting of convergence issues can be done by changing input in this folder

After running the simulation you can see the results of your simulation in this folder

Example 1.1 and 1.2



Reference

- AspenTech EAP101 Introduction to Process Modeling using Aspen Plus
- Warren D. Seider, J. D. Seader, Daniel R. Lewin, Product and Process Design Principles: Synthesis, Analysis and Design, Third Edition, Wiley, 2008.